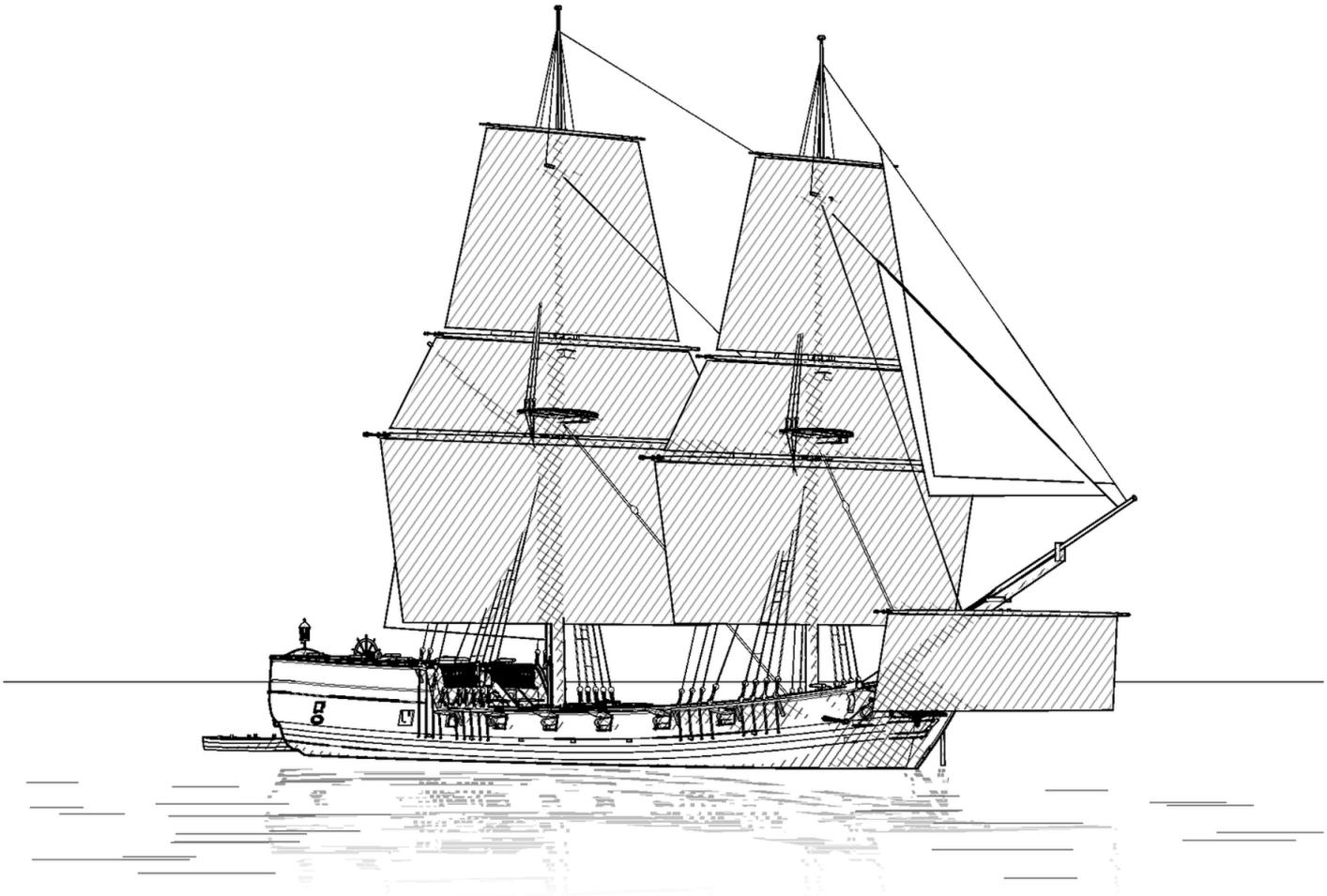
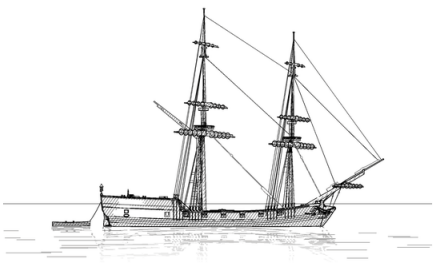
How to Draw a Sailing Ship

Step by step, in three views

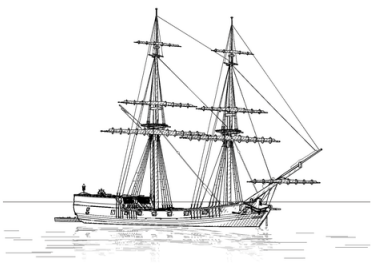


Our snow: a small two-masted naval ship of about 1710

For curious artists aged about 12 to 16 (and anyone else).



from the side



from the bow



from the stern

Every picture in this booklet was made by a computer script from the 3D model of our reference ship: the lines are traced from the model, not drawn by hand.

MIT License. Drawings rendered as line art from the 3D model of the reference ship.

Before you start

What you need

- Plain paper, A4 or bigger, lying sideways (landscape)
- A hard pencil (H or HB) for light construction lines
- A soft pencil (2B or 4B) for the final lines and shading
- A ruler, best a long one (30 cm)
- A soft eraser, and a kneaded one if you have it
- Optional: a black fine-liner pen for inking

How to use this booklet

- Pick one of the three views. The side view is the easiest; start there if this is your first ship.
- Each step adds one group of things. Draw only what is new in that step.
- Draw construction lines very lightly. You will erase them at the end.
- Copy proportions, not millimetres: compare lengths with each other.
- Take your time. A ship takes two or three evenings, and that is normal.

Colours in the step pictures

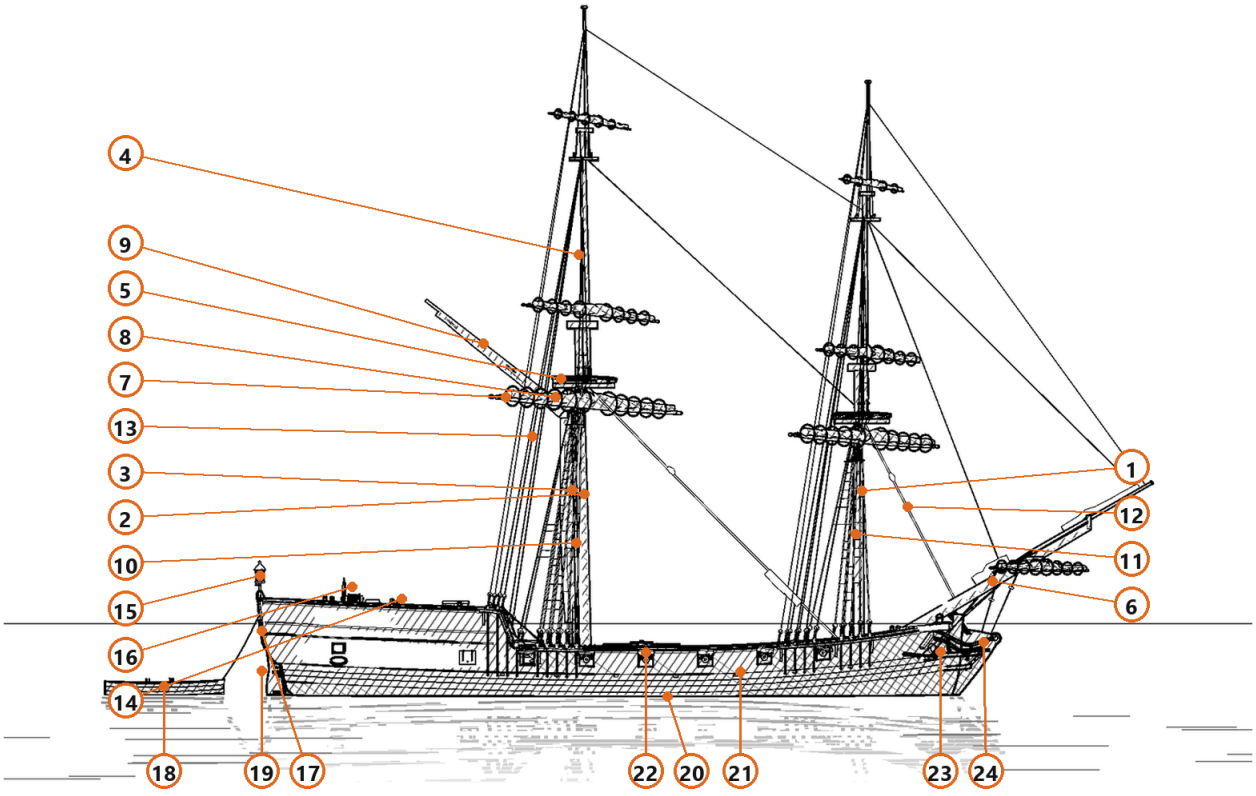
black: new lines for this step

light blue: lines you drew in earlier steps

orange dashes: construction lines (draw lightly, erase later)

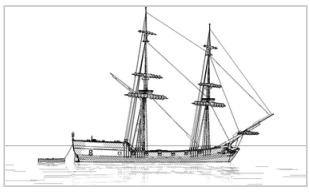
Meet the ship

These names come up in the steps. You do not have to learn them all, but they help you know what you are drawing.



1. foremast
2. mainmast
3. snow mast (for the gaff sail)
4. topmast
5. top (platform)
6. bowsprit
7. yard
8. furled sail
9. gaff
10. shrouds
11. ratlines (rope steps)
12. stay
13. backstay
14. quarterdeck
15. stern lantern
16. steering wheel
17. stern with windows
18. ship's boat
19. rudder
20. waterline
21. wale
22. gunport
23. anchor
24. head (bow decoration)

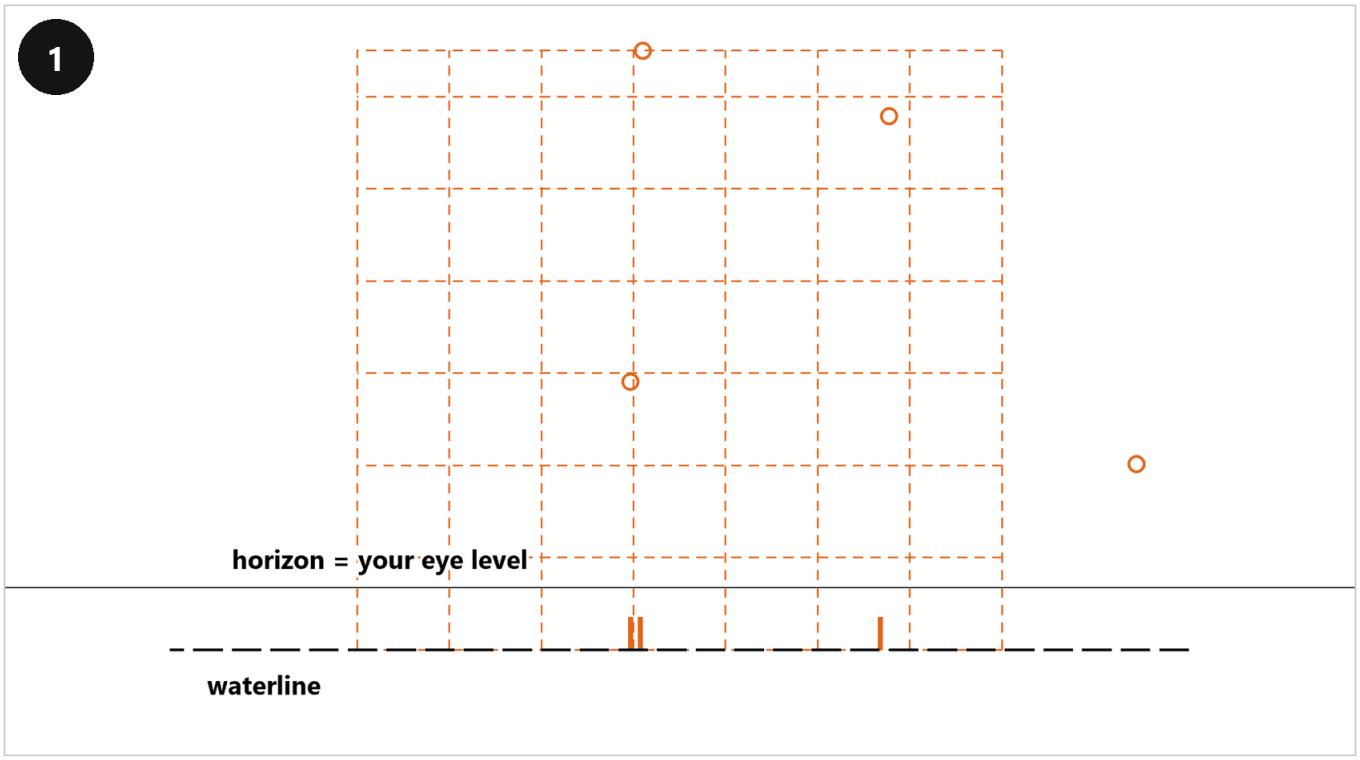
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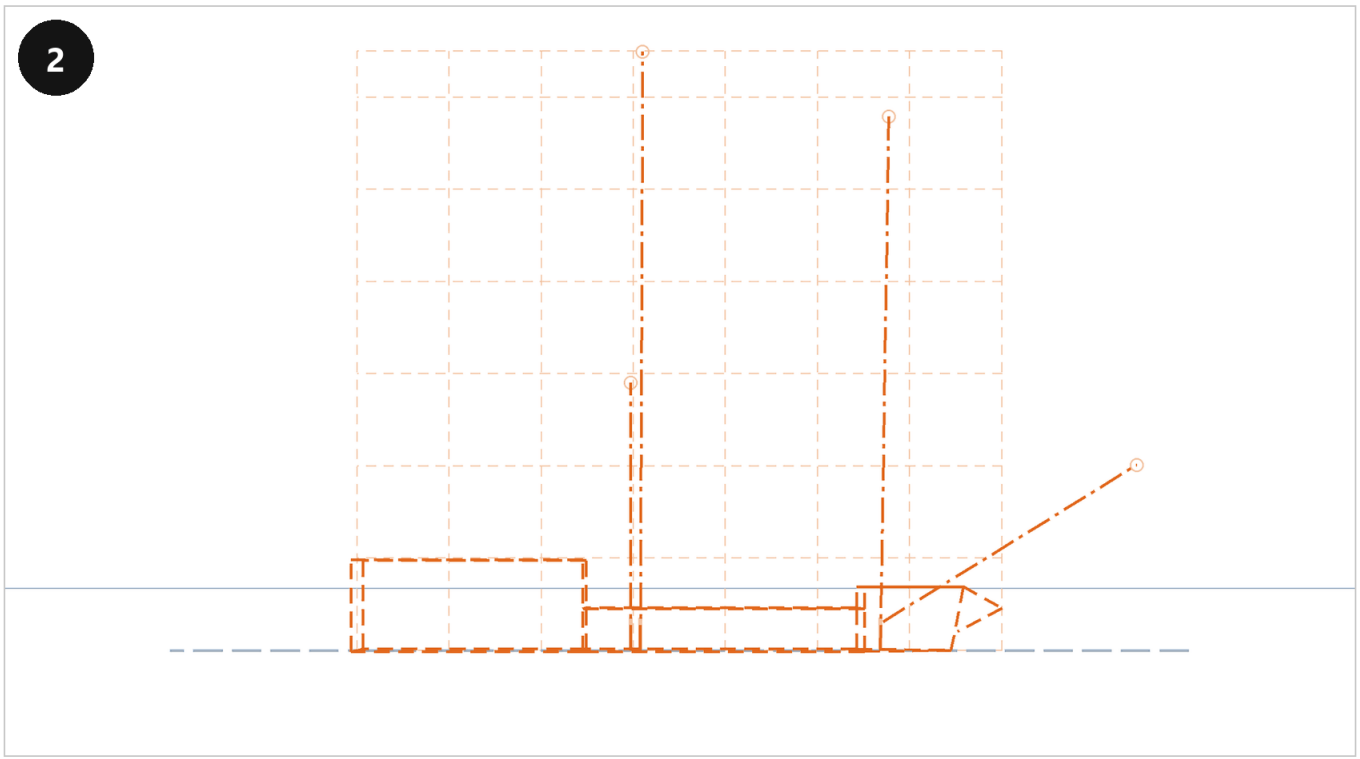
View 1: from the side

Difficulty: ●○○ **Easy**

A flat side view, like a plan. Good for learning the proportions.

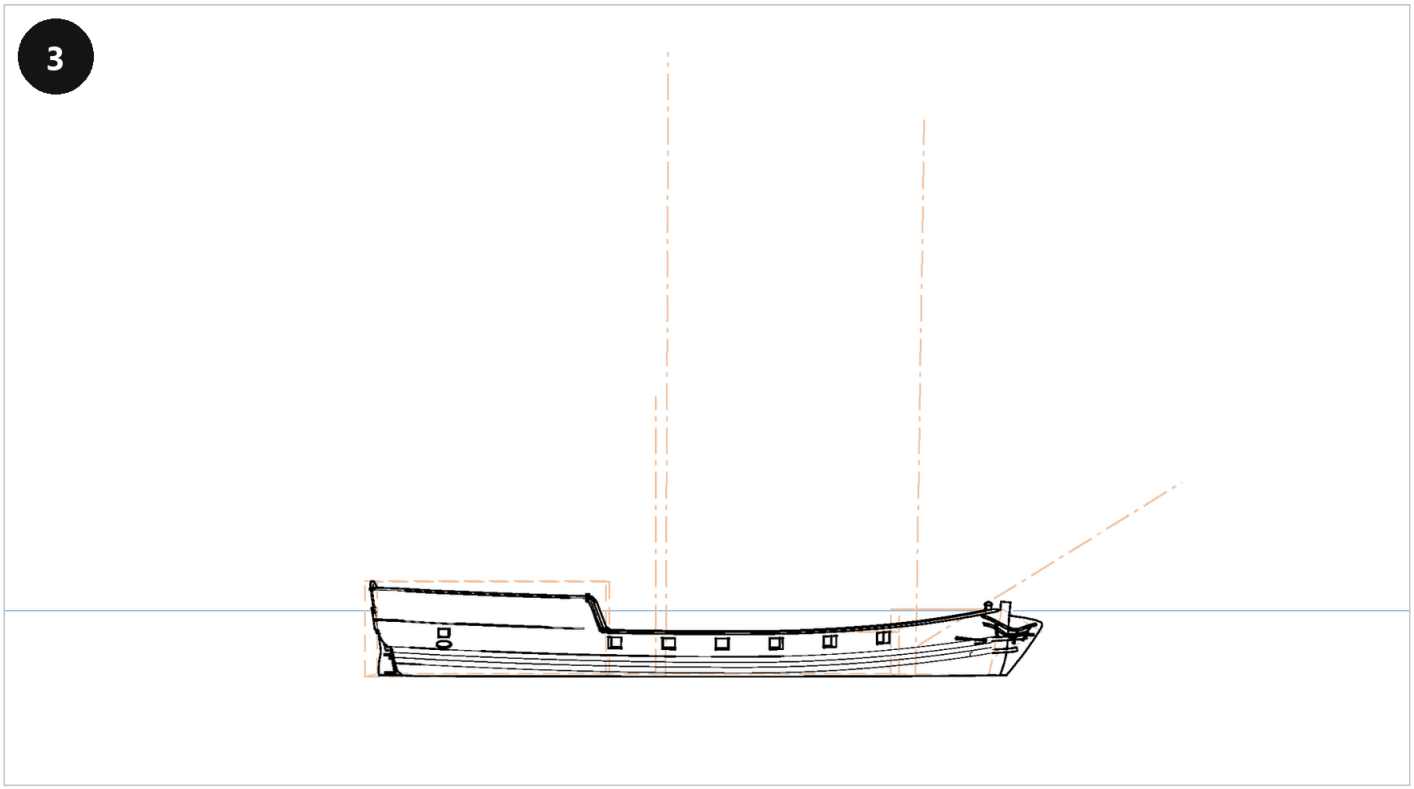


Step 1. Horizon, water and a grid. Draw the horizon across the paper: it is at the height of your eyes. Just below it goes the waterline. On the waterline draw a grid of 7 equal squares for the length of the hull and 6½ squares up. Mark the masts: the foremast about 1¼ squares from the bow, the mainmast about 4. Their tops reach about 5¾ and 6½ squares up.

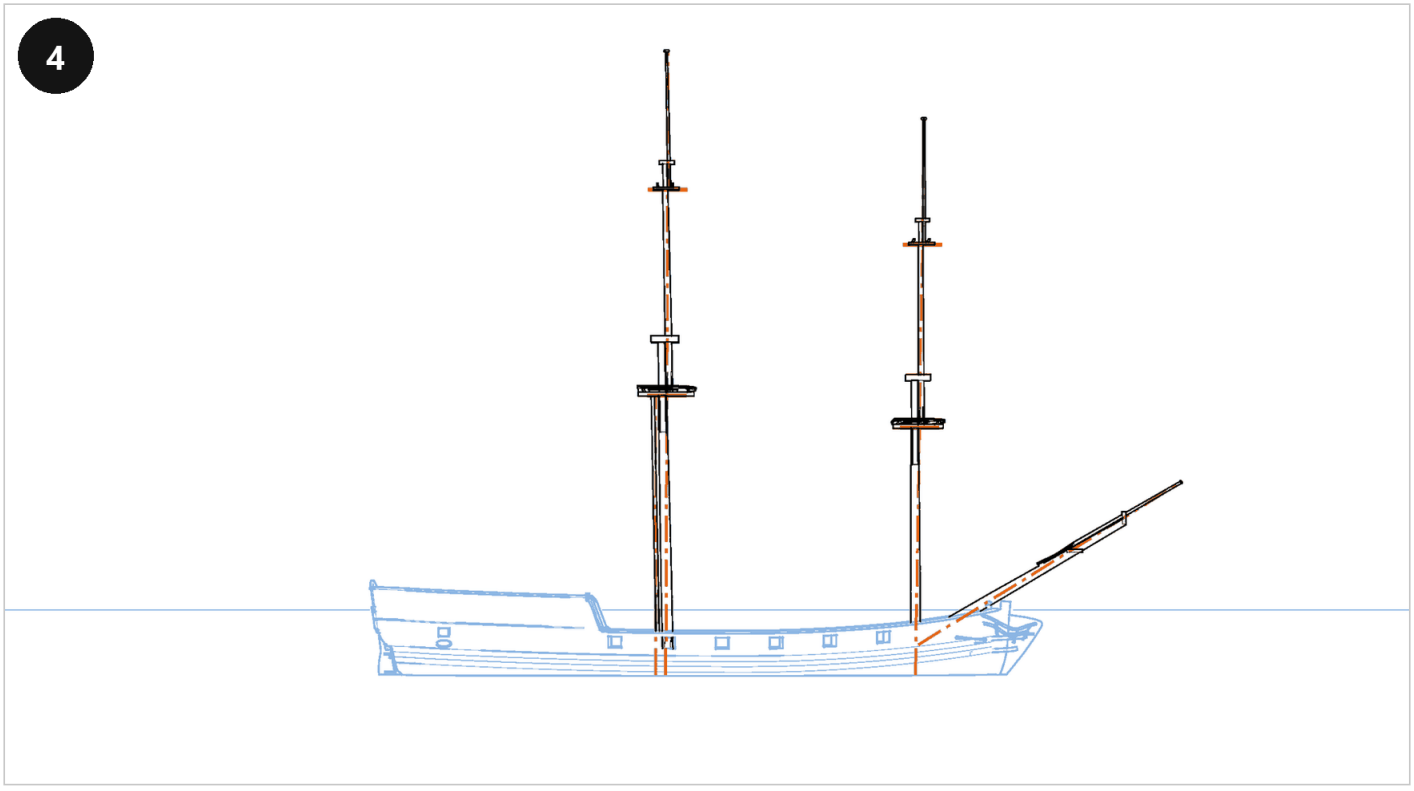


Step 2. Simple blocks. Build the hull from simple shapes: a long low box in the middle, a taller box at the stern (the quarterdeck), and a wedge at the bow. Then draw the mast axes straight up from your marks, and a slanting line for the bowsprit, about 1½ squares beyond the bow.

MIT License. Drawings rendered as line art from the 3D model of the reference ship.

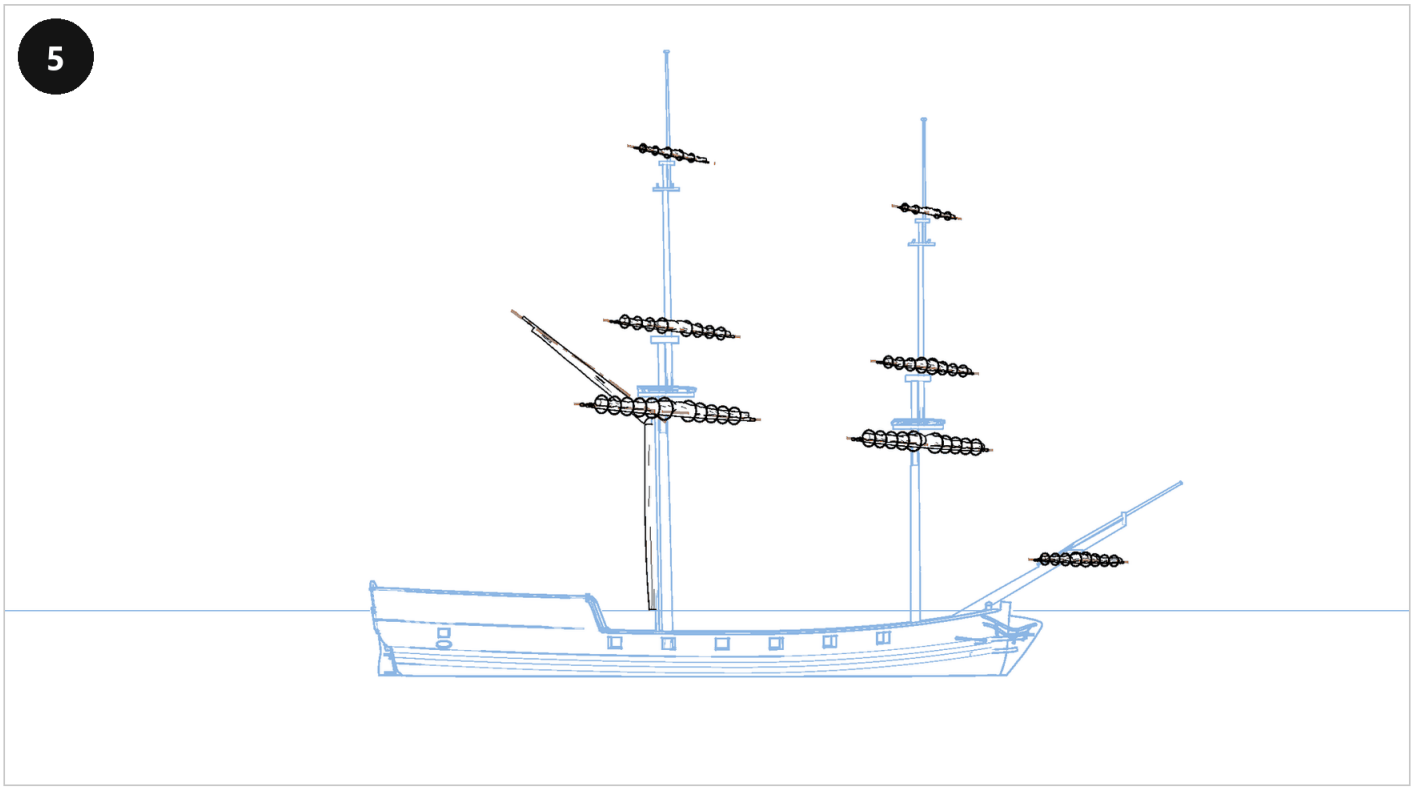


Step 3. The hull outline. Now draw the real shape of the hull over your blocks. Start with the waterline, then the line of the rail: it is not straight but rises a little towards the bow and steps up at the quarterdeck. Add the stem at the bow, the stern, the rudder and the long lines of the wales along the side.

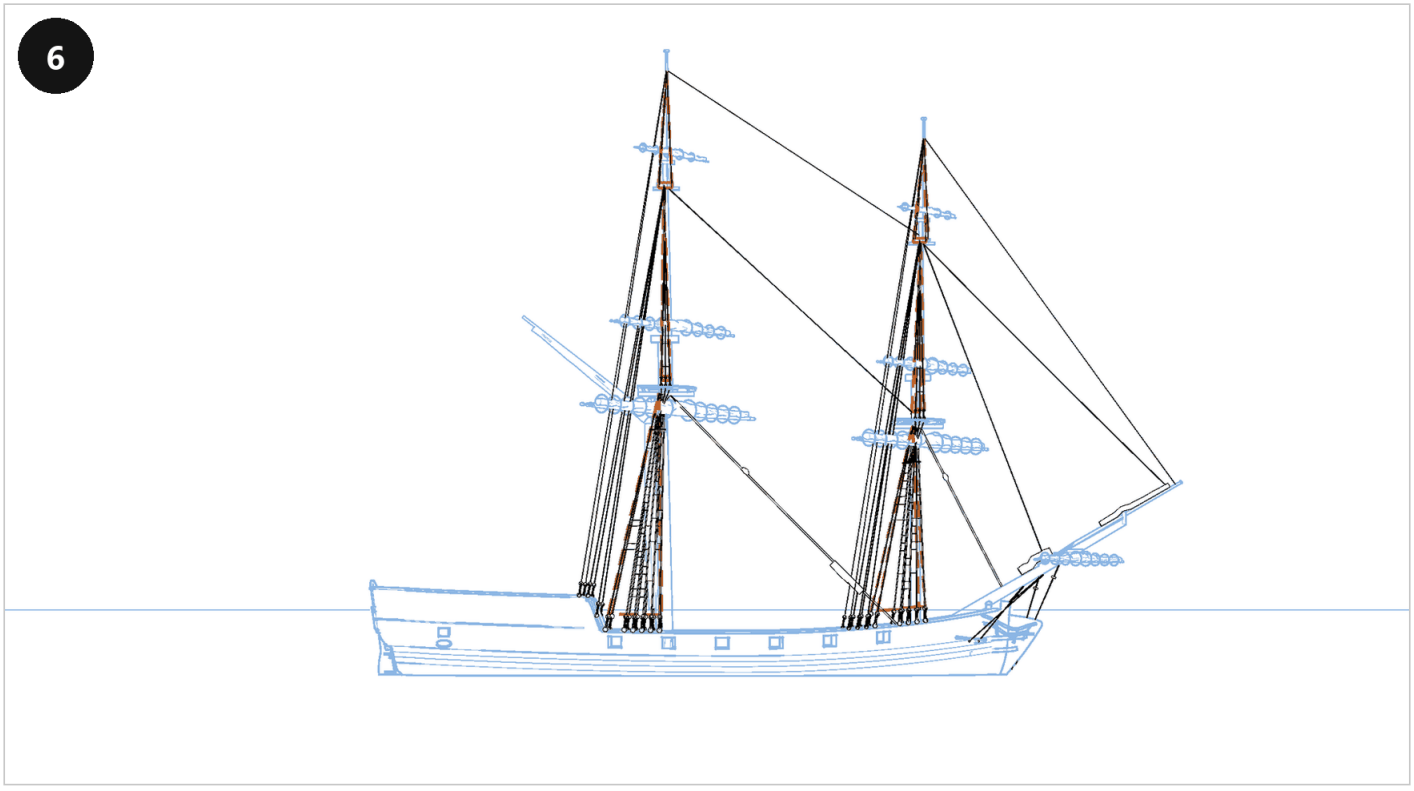


Step 4. Masts and bowsprit. Give the mast lines their thickness: each mast gets thinner towards the top and is made of three pieces (orange crosses show where one piece ends). Draw the tops, the small platforms where the pieces meet. Then the bowsprit, and the thin snow mast behind the mainmast.

MIT License. Drawings rendered as line art from the 3D model of the reference ship.

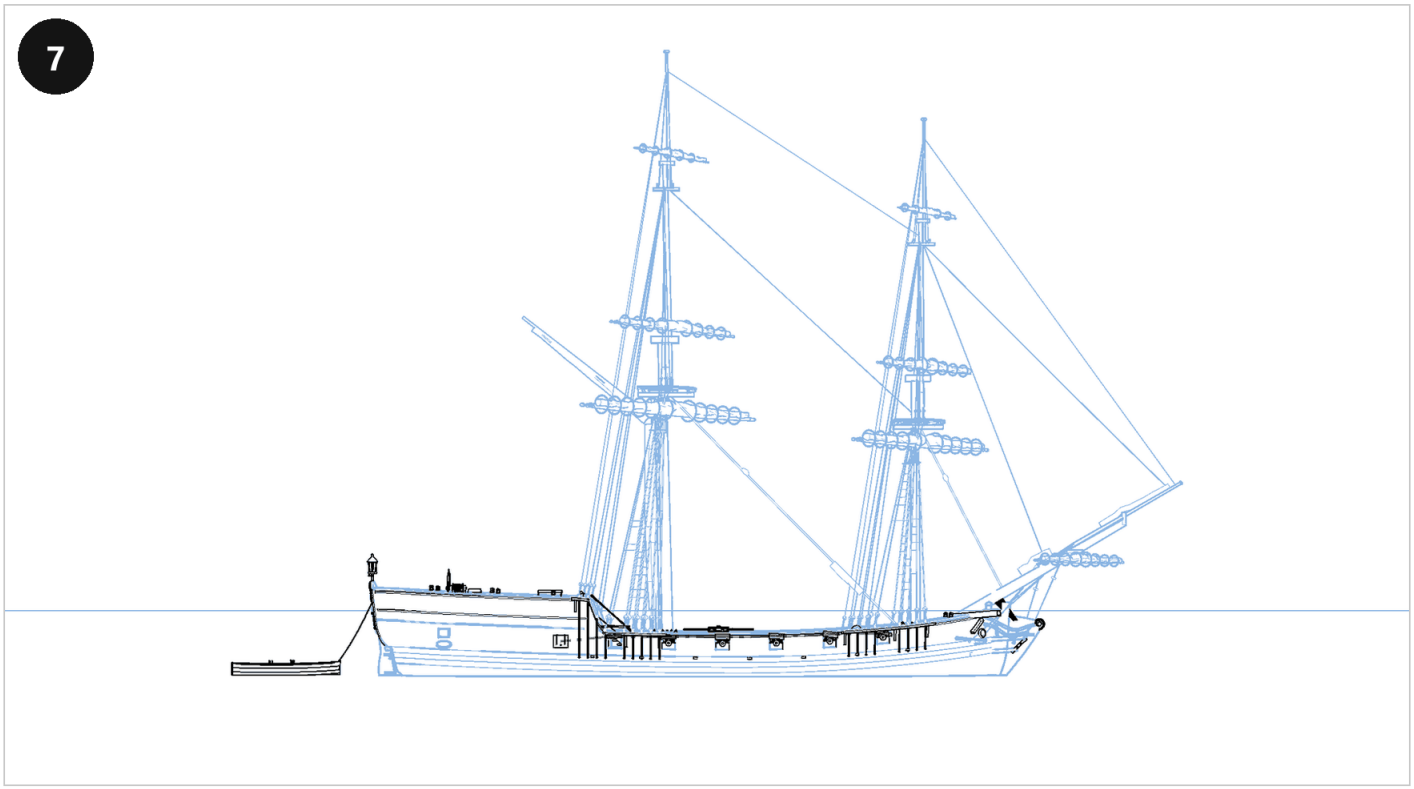


Step 5. Yards, furled sails and the gaff. Draw each yard as a long thin shape crossing its mast, using the orange centre lines. The rolled-up sails sit on top of the yards like sausages. Add the gaff: the slanting spar behind the mainmast. From the side the yards point a little towards you, so they look short.

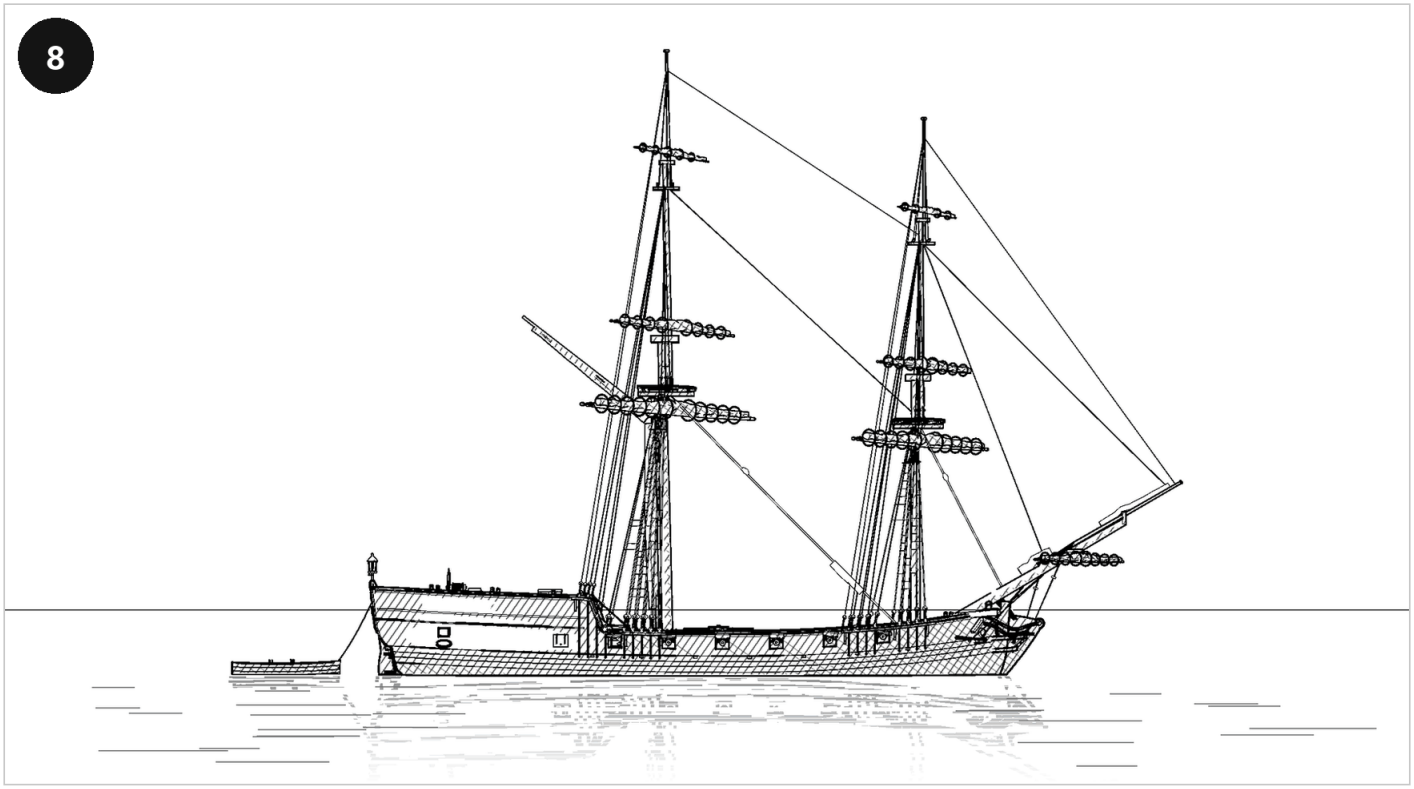


Step 6. Standing rigging. Stays first: long straight lines running forward and down from each mast. Then the shrouds: draw the orange triangles lightly and fill them with lines fanning down to the side of the hull. Ratlines last, and only a few: every third one is plenty.

MIT License. Drawings rendered as line art from the 3D model of the reference ship.

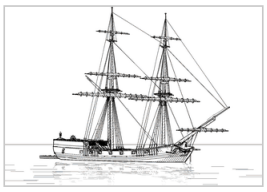


Step 7. Hull details. Add the gunport lids along the orange line, with a gun barrel in most of the ports, and the channels: the ledges on the side where the shrouds end. Then the small things: windows and lantern at the stern, anchor at the bow, the steering wheel and the boat on its line behind the ship.



Step 8. Shading, reflection, finish. Go over the lines you want to keep with a soft pencil or a pen, and erase all construction lines. Shade the side of the hull that is away from the light with close parallel lines. Draw the reflection as broken, wobbly strokes under the ship.

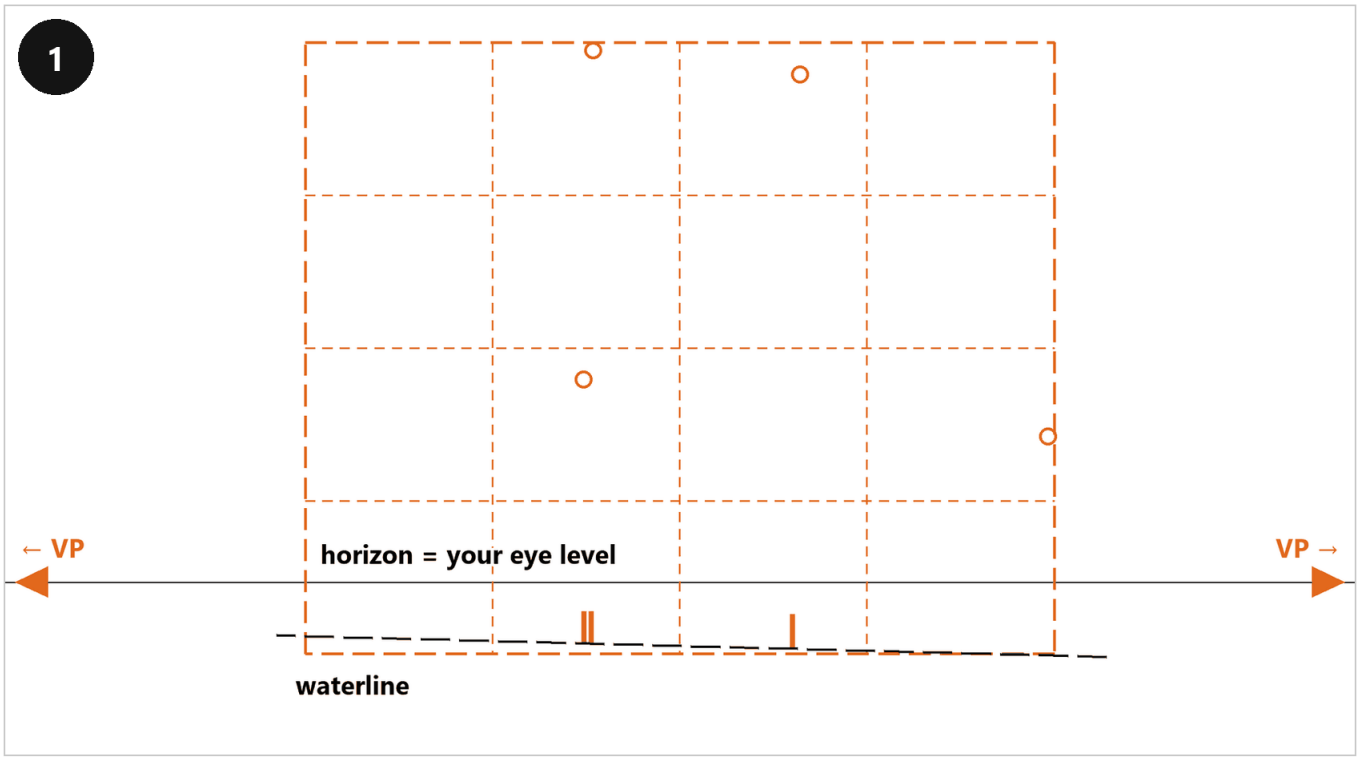
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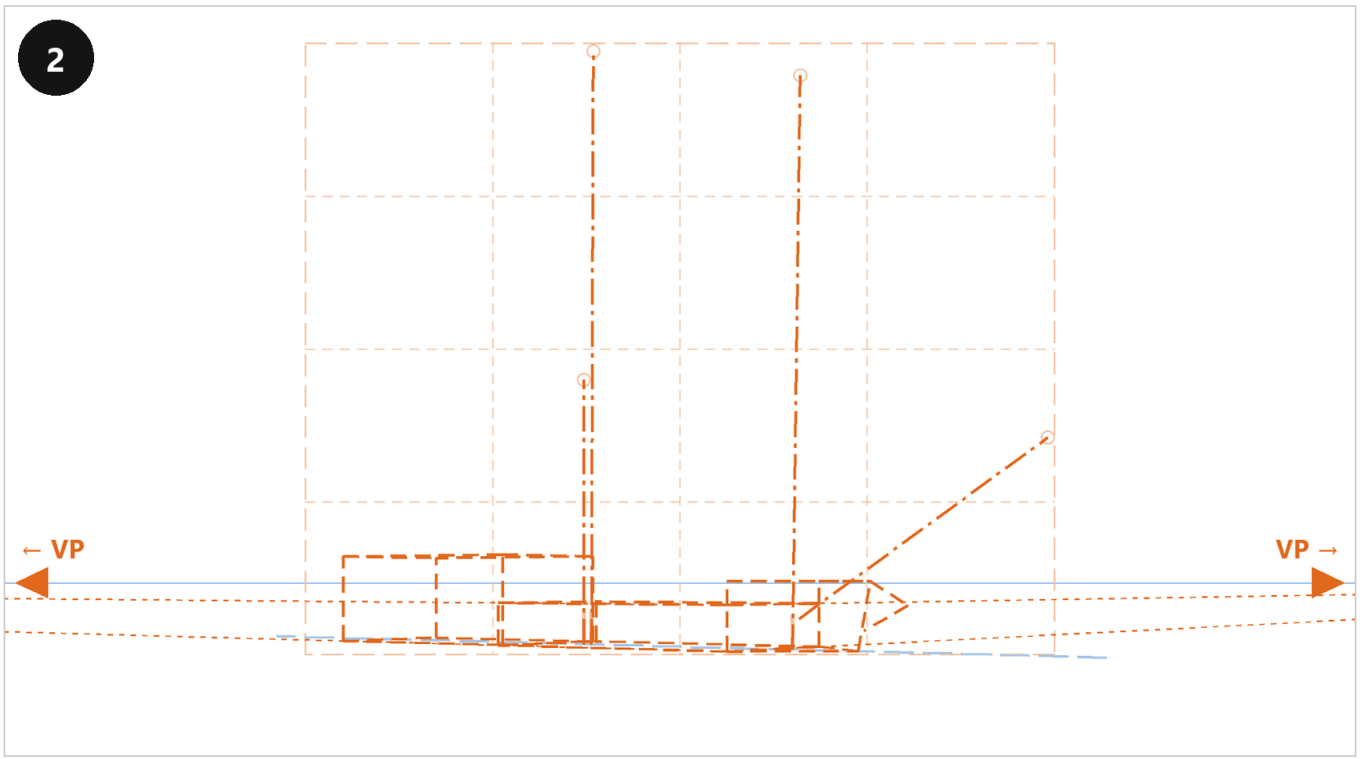
View 2: from the bow

Difficulty: ●●○ **Medium**

Seen at an angle from the front. You will use vanishing points.

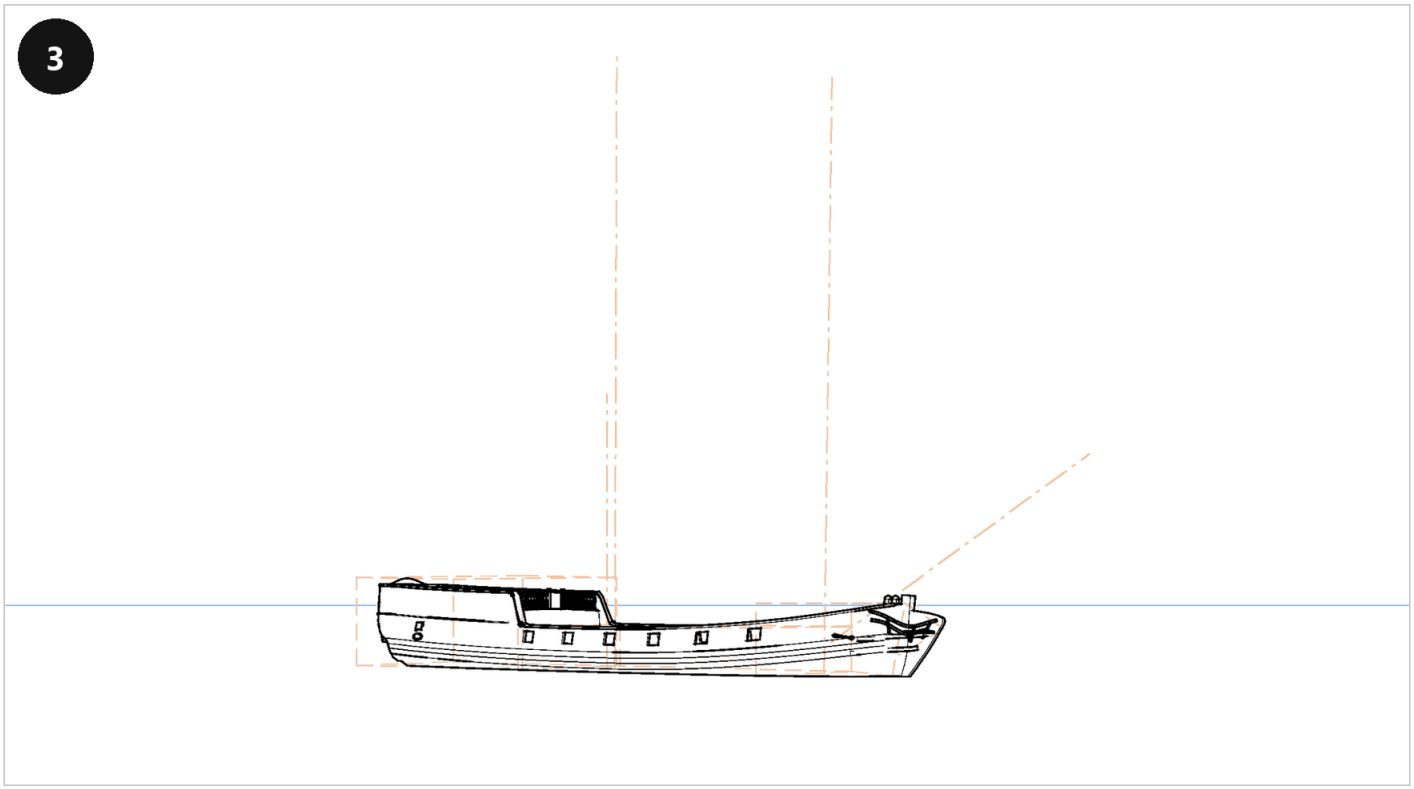


Step 1. Horizon, water and a frame. Draw the horizon at your eye level, low on the page. Both vanishing points (VP) lie on it, far outside the paper to the left and right. Draw the waterline slanting gently towards one of them. Then a light frame for the whole ship, divided into 4 by 4, and mark where the masts stand and how high they reach.

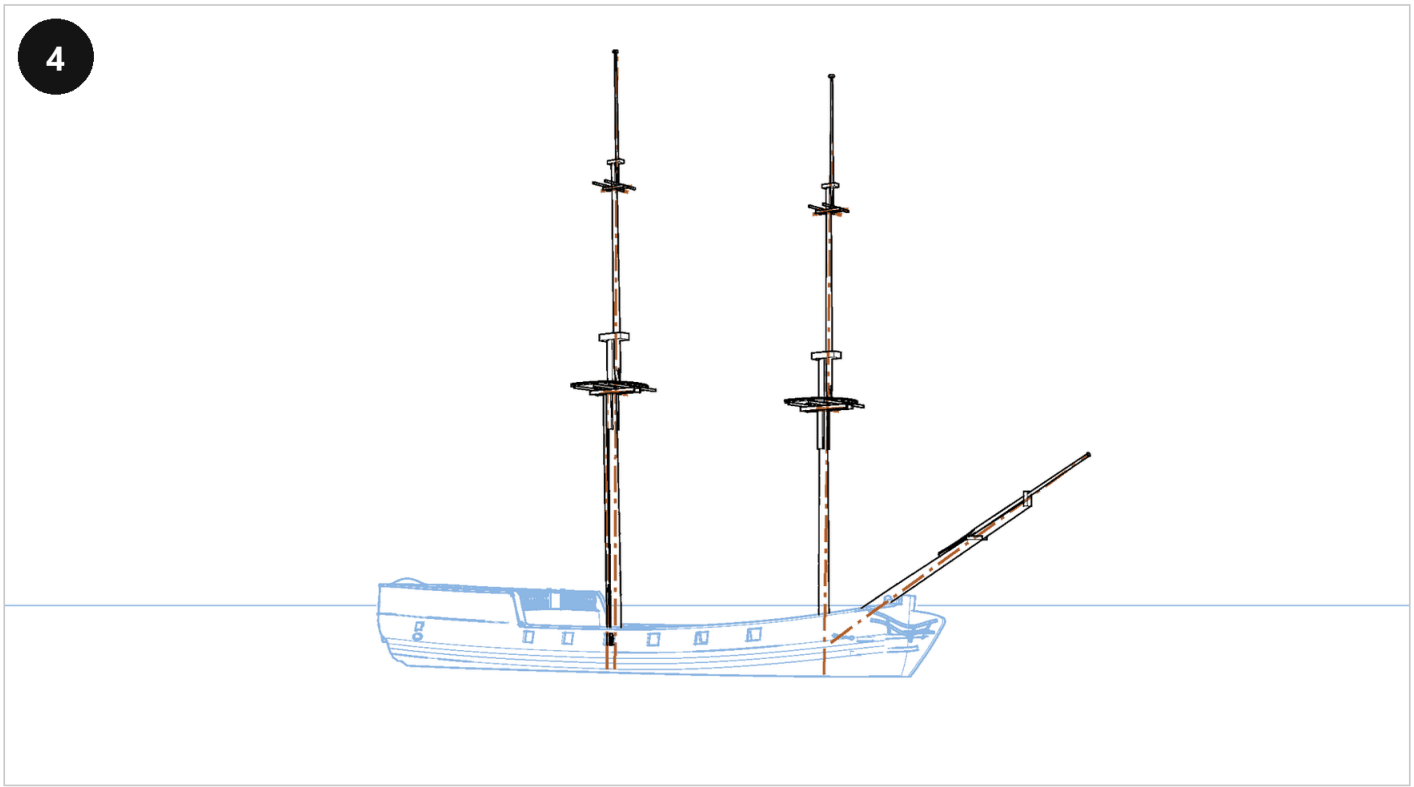


Step 2. Boxes in perspective. Draw the hull as boxes in perspective. Lines along the ship run towards one vanishing point, lines across it towards the other; upright lines stay upright. Use a ruler and aim at your VP marks. Add the bow wedge, then the mast axes and the bowsprit line.

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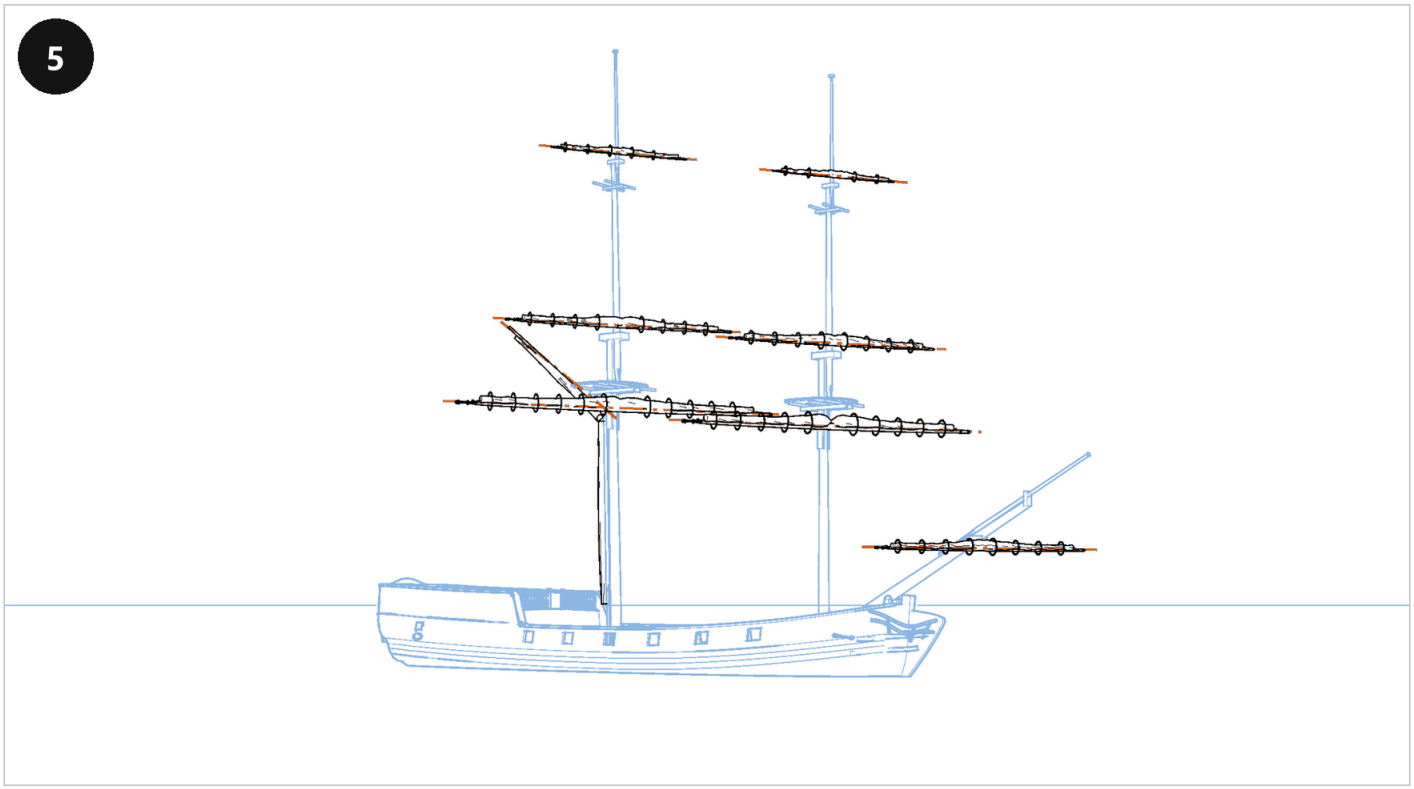


Step 3. The hull outline. Now draw the real shape of the hull over your blocks. Start with the waterline, then the line of the rail: it is not straight but rises a little towards the bow and steps up at the quarterdeck. Add the stem at the bow, the stern, the rudder and the long lines of the wales along the side. At the bow the hull narrows to the stem; the head with its decoration sticks out in front.

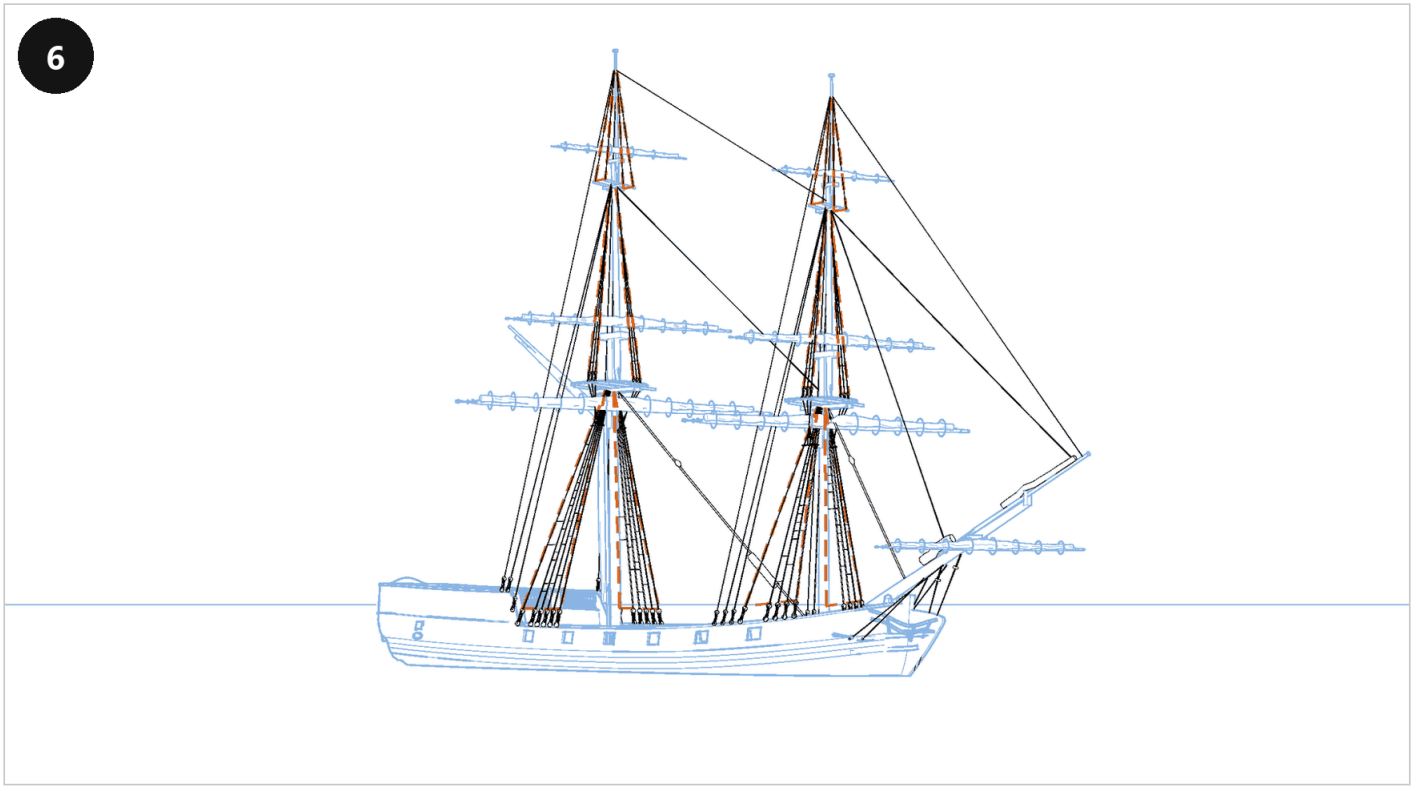


Step 4. Masts and bowsprit. Give the mast lines their thickness: each mast gets thinner towards the top and is made of three pieces (orange crosses show where one piece ends). Draw the tops, the small platforms where the pieces meet. Then the bowsprit, and the thin snow mast behind the mainmast.

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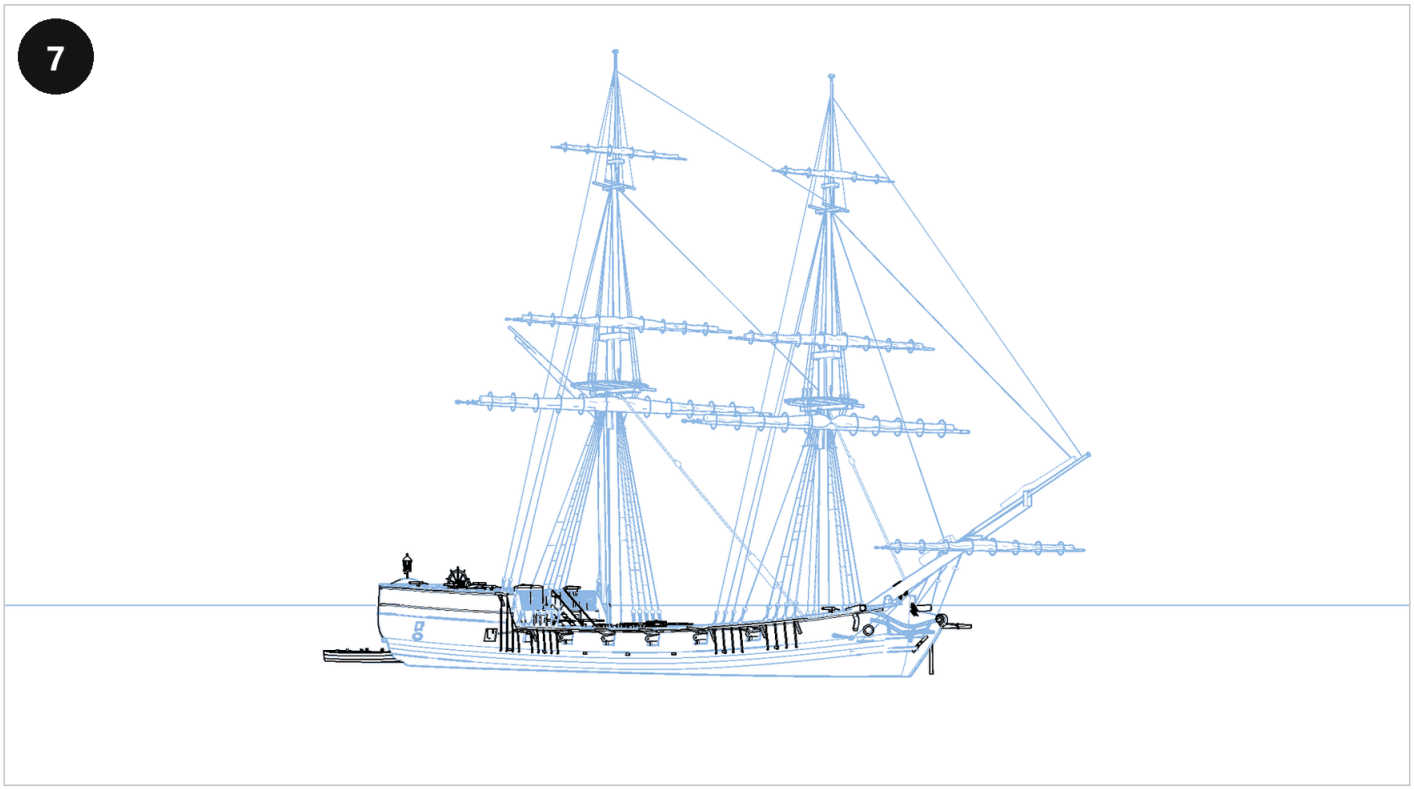


Step 5. Yards, furled sails and the gaff. Draw each yard as a long thin shape crossing its mast, using the orange centre lines. The rolled-up sails sit on top of the yards like sausages. Add the gaff: the slanting spar behind the mainmast. The yards also follow the perspective: the far ends look shorter.

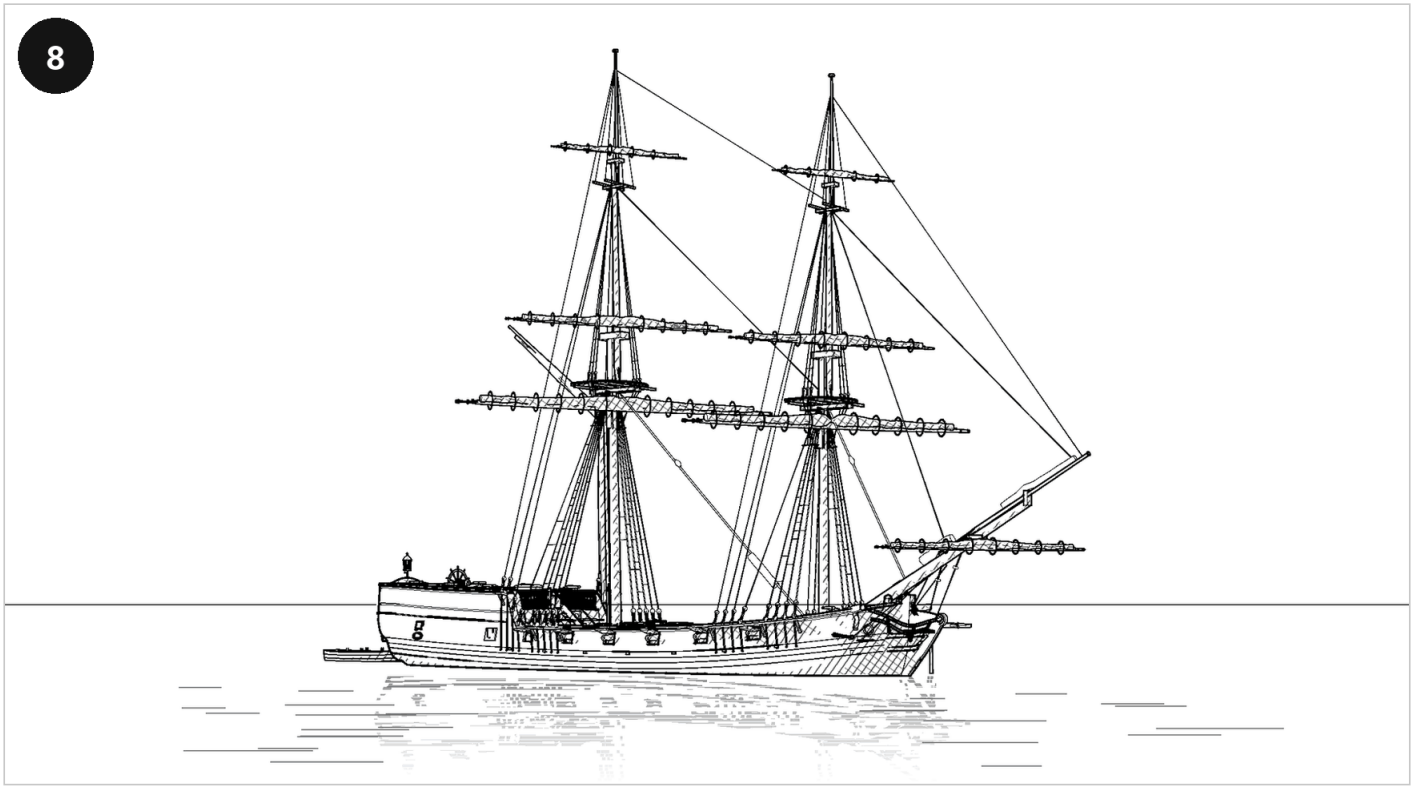


Step 6. Standing rigging. Stays first: long straight lines running forward and down from each mast. Then the shrouds: draw the orange triangles lightly and fill them with lines fanning down to the side of the hull. Ratlines last, and only a few: every third one is plenty.

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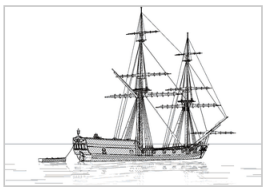


Step 7. Hull details. Add the gunport lids along the orange line, with a gun barrel in most of the ports, and the channels: the ledges on the side where the shrouds end. Then the small things: windows and lantern at the stern, anchor at the bow, the steering wheel and the boat on its line behind the ship.



Step 8. Shading, reflection, finish. Go over the lines you want to keep with a soft pencil or a pen, and erase all construction lines. Shade the side of the hull that is away from the light with close parallel lines. Draw the reflection as broken, wobbly strokes under the ship.

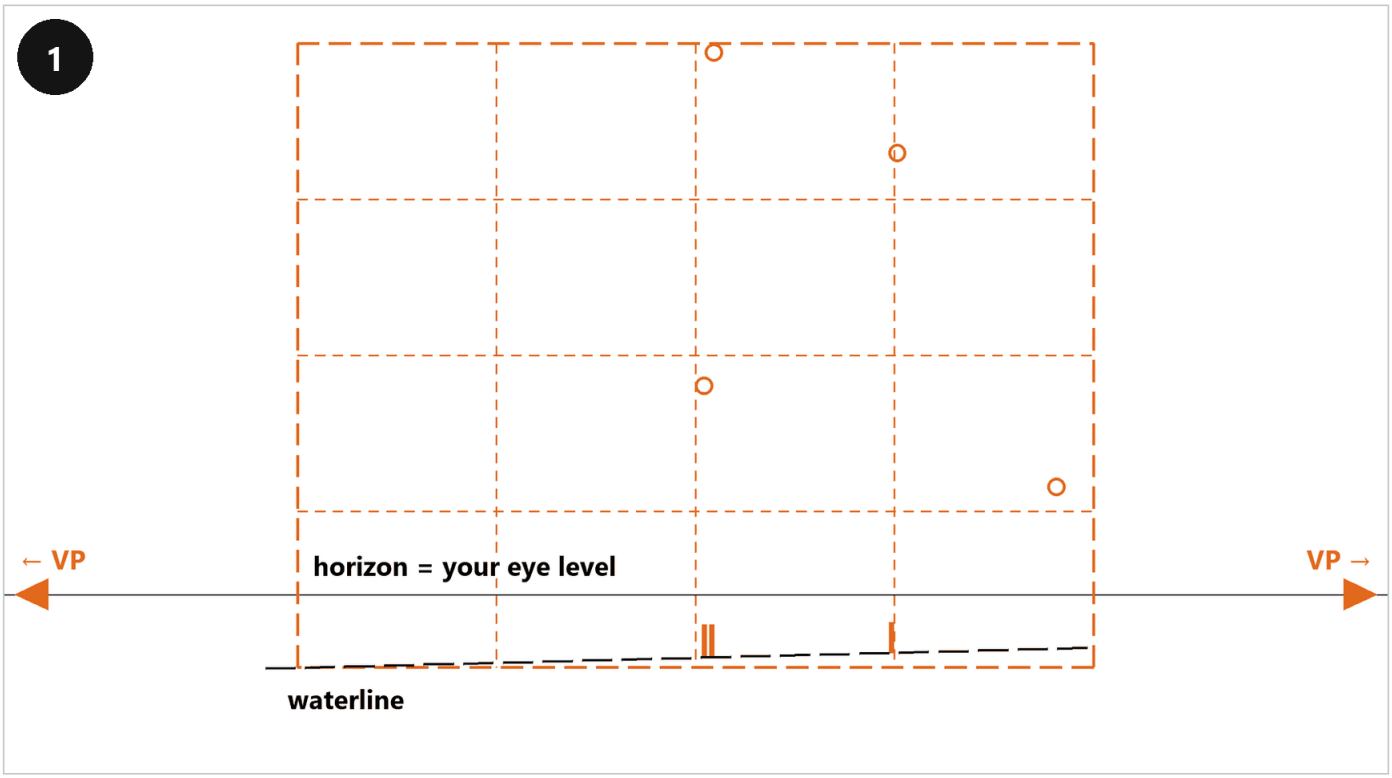
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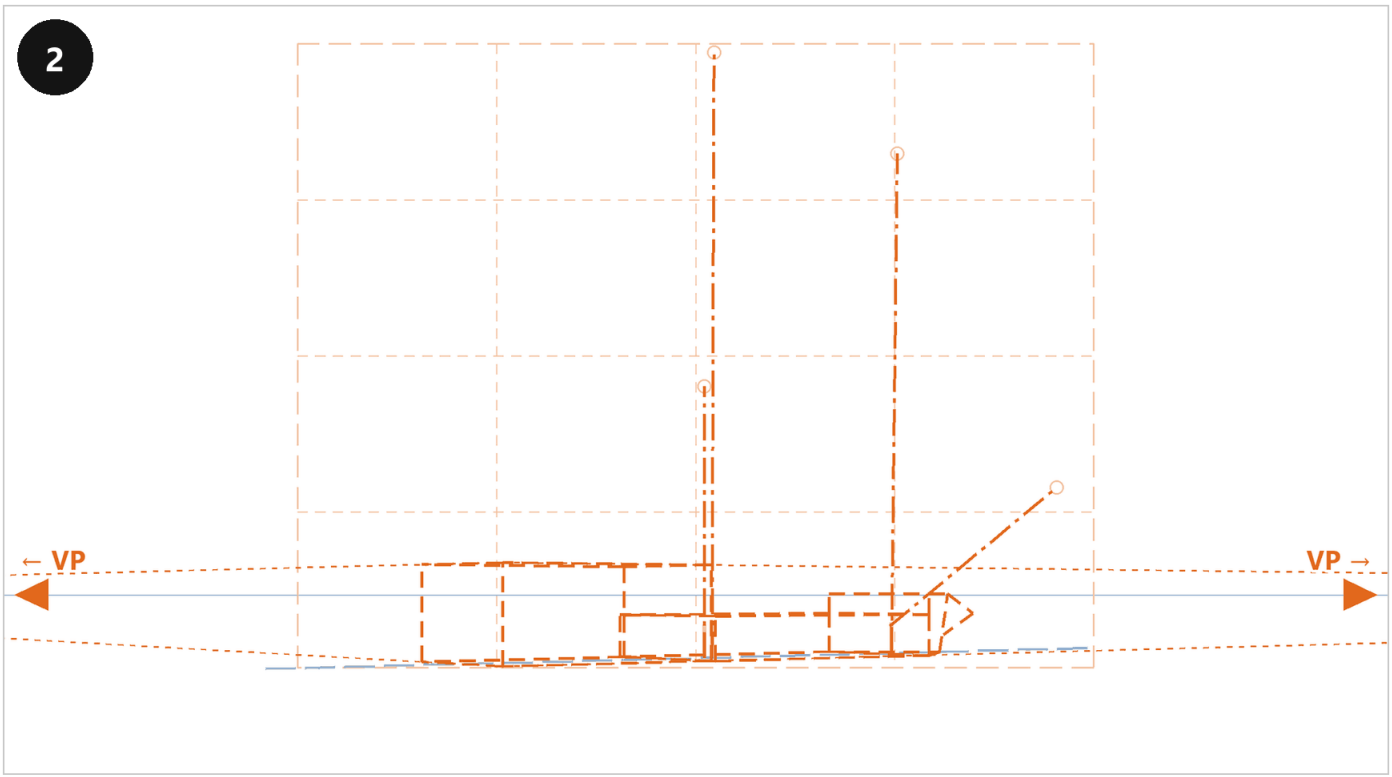
View 3: from the stern

Difficulty: ●●● **Harder**

Seen at an angle from behind: decorated stern, boat, more overlaps.

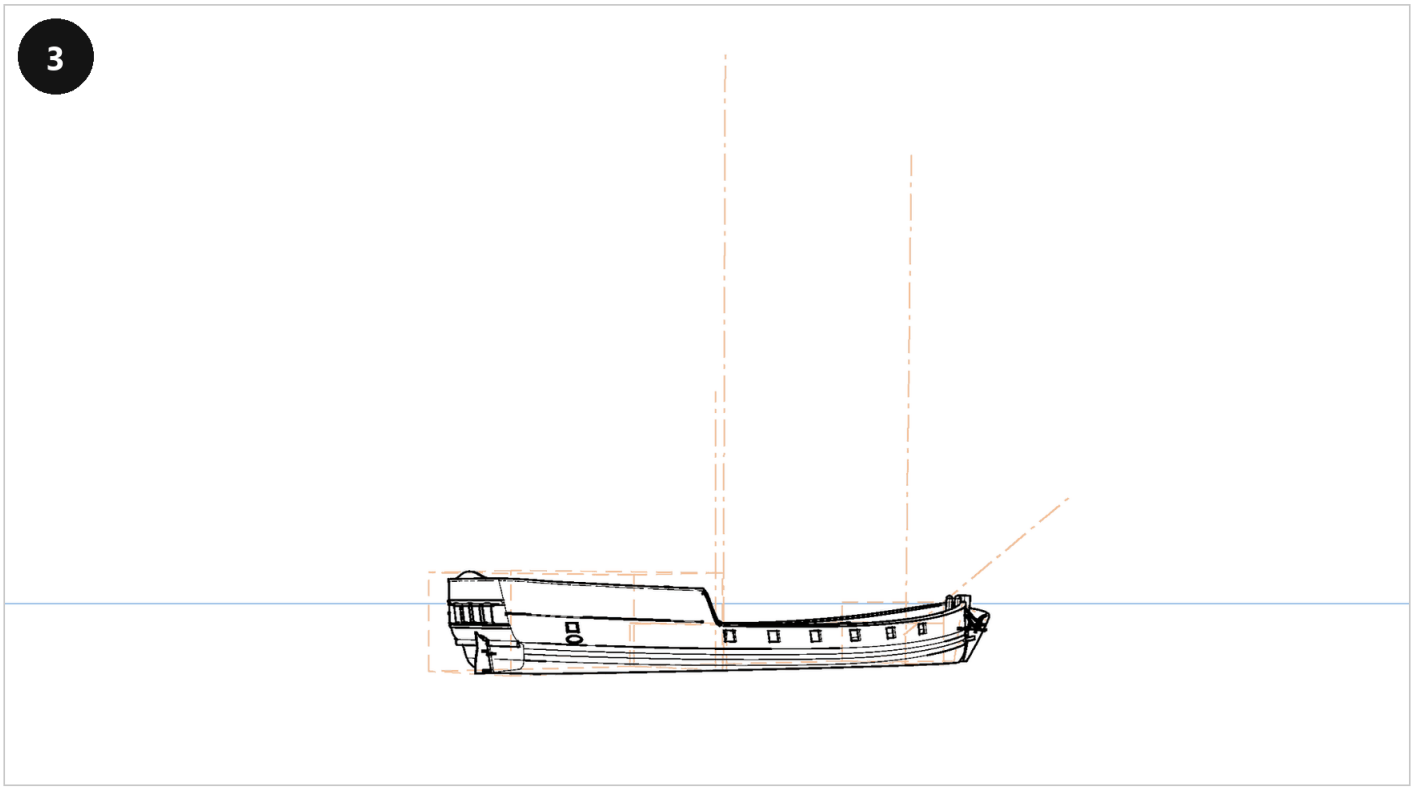


Step 1. Horizon, water and a frame. Draw the horizon at your eye level. The vanishing points (VP) are on it, outside the paper. Draw the waterline sloping gently towards one of them, then a light frame divided into 4 by 4. Mark the mast positions and tops: here the mainmast is the one closer to you, so it looks taller.

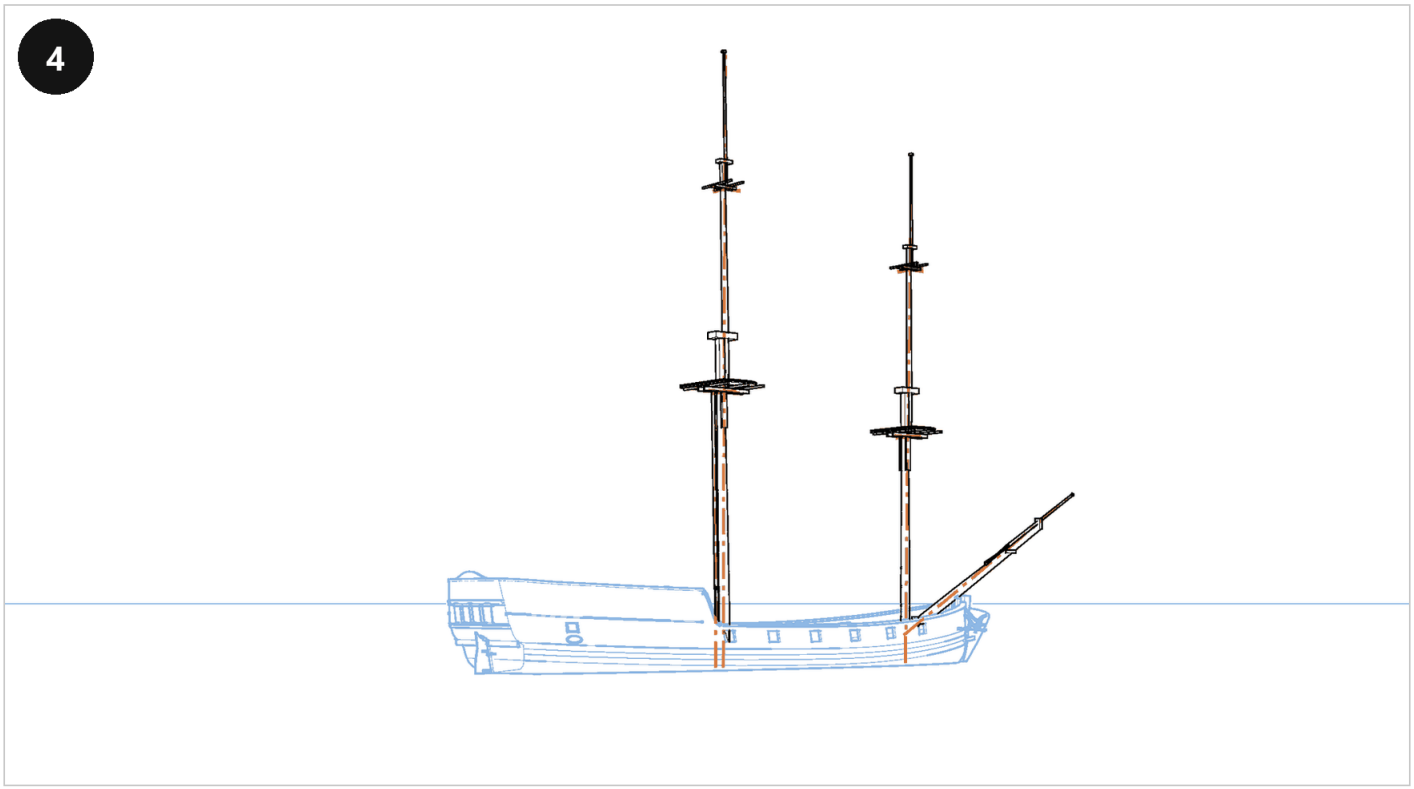


Step 2. Boxes in perspective. The stern is a tall box facing you at an angle, the rest of the hull a longer, lower box behind it. Aim every line along the ship at one VP and every line across it at the other. Then draw the mast axes and the bowsprit line.

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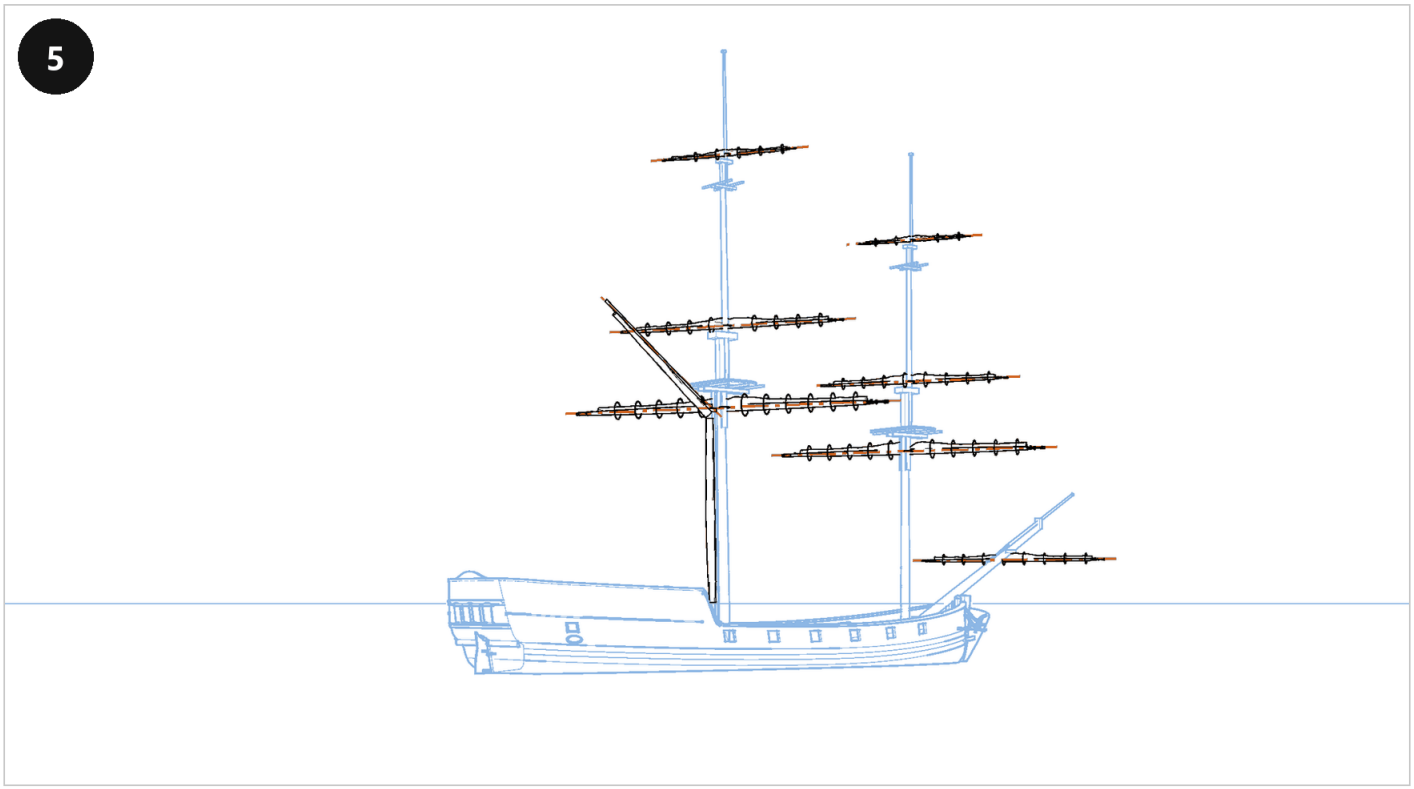


Step 3. The hull outline. Now draw the real shape of the hull over your blocks. Start with the waterline, then the line of the rail: it is not straight but rises a little towards the bow and steps up at the quarterdeck. Add the stem at the bow, the stern, the rudder and the long lines of the wales along the side. The stern is wide and flat with windows; draw its outline first, the windows come in step 7.

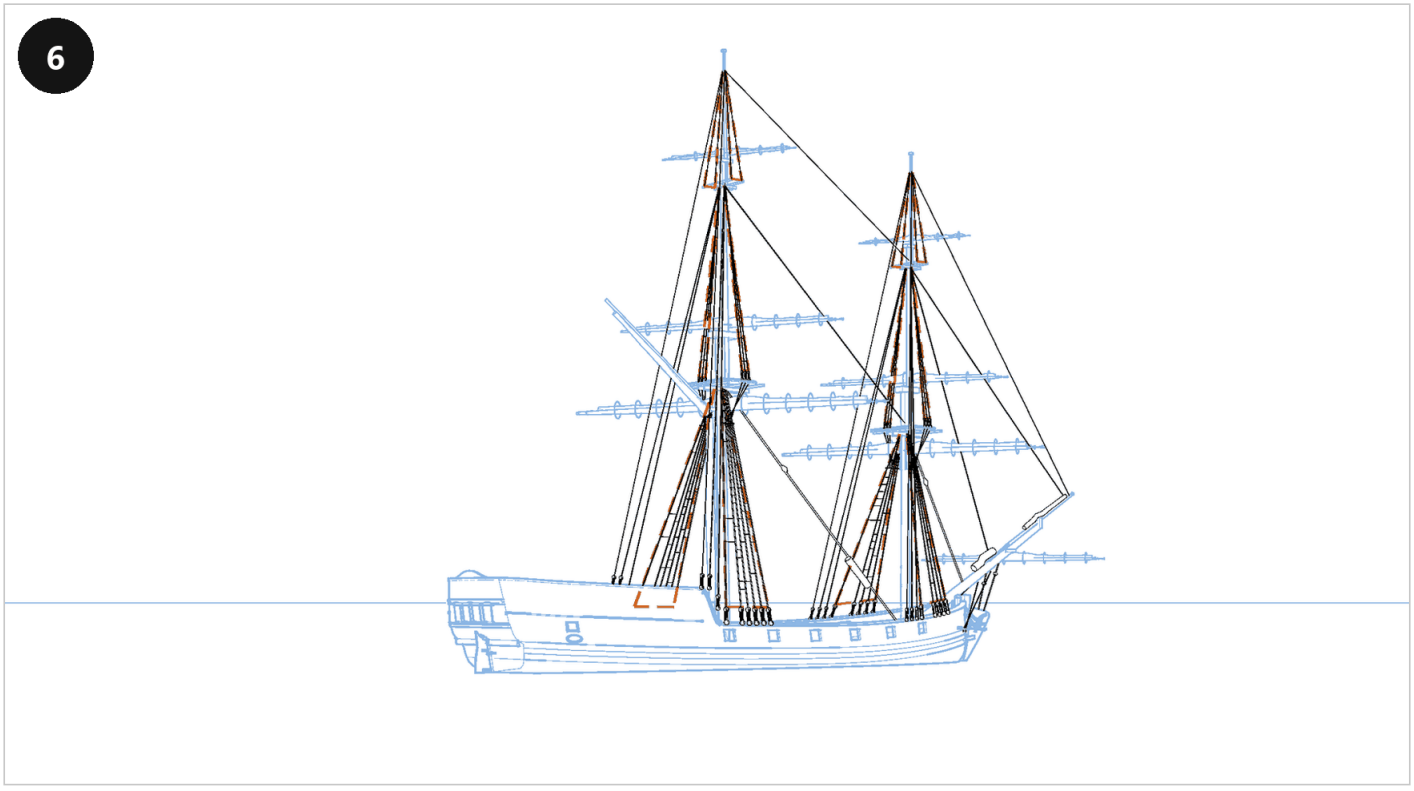


Step 4. Masts and bowsprit. Give the mast lines their thickness: each mast gets thinner towards the top and is made of three pieces (orange crosses show where one piece ends). Draw the tops, the small platforms where the pieces meet. Then the bowsprit, and the thin snow mast behind the mainmast.

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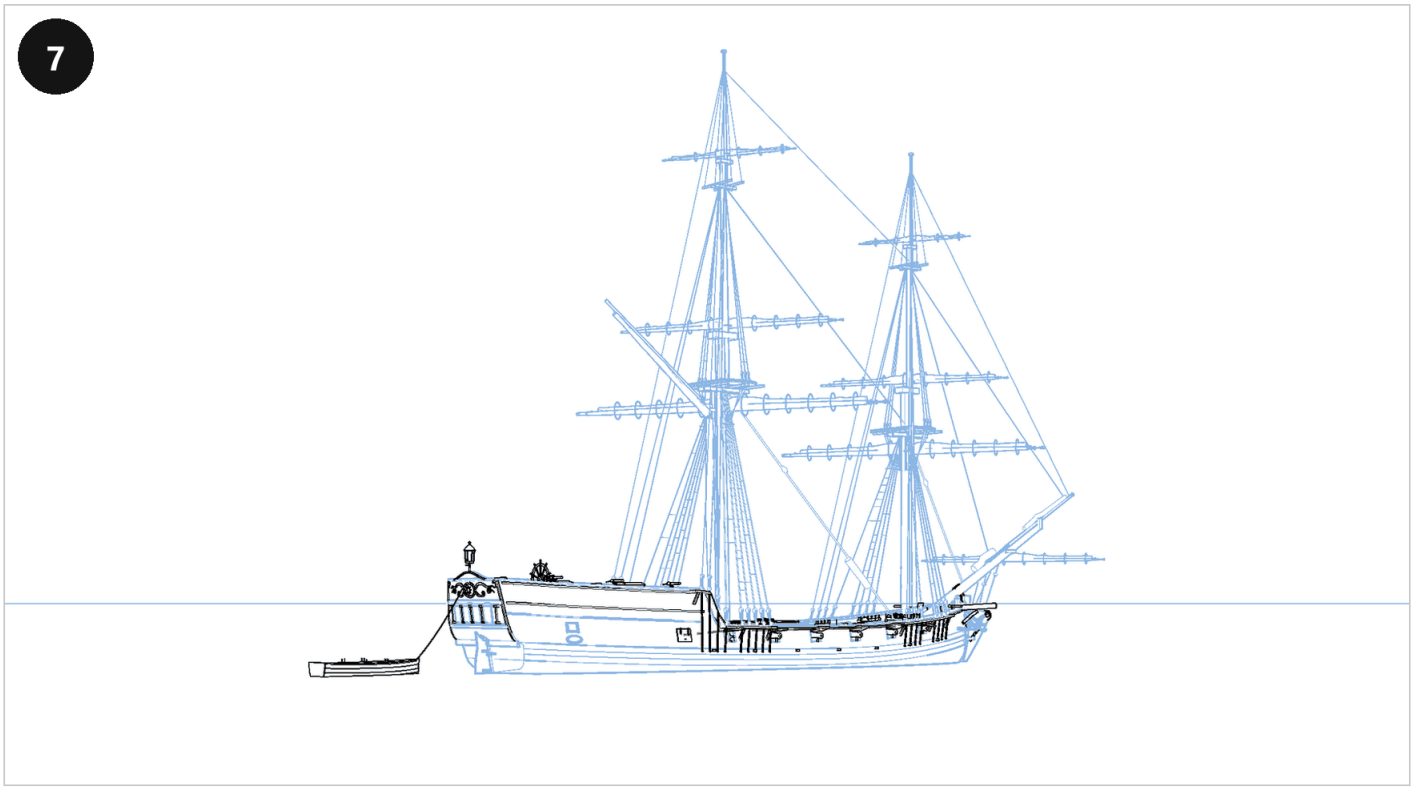


Step 5. Yards, furled sails and the gaff. Draw each yard as a long thin shape crossing its mast, using the orange centre lines. The rolled-up sails sit on top of the yards like sausages. Add the gaff: the slanting spar behind the mainmast.

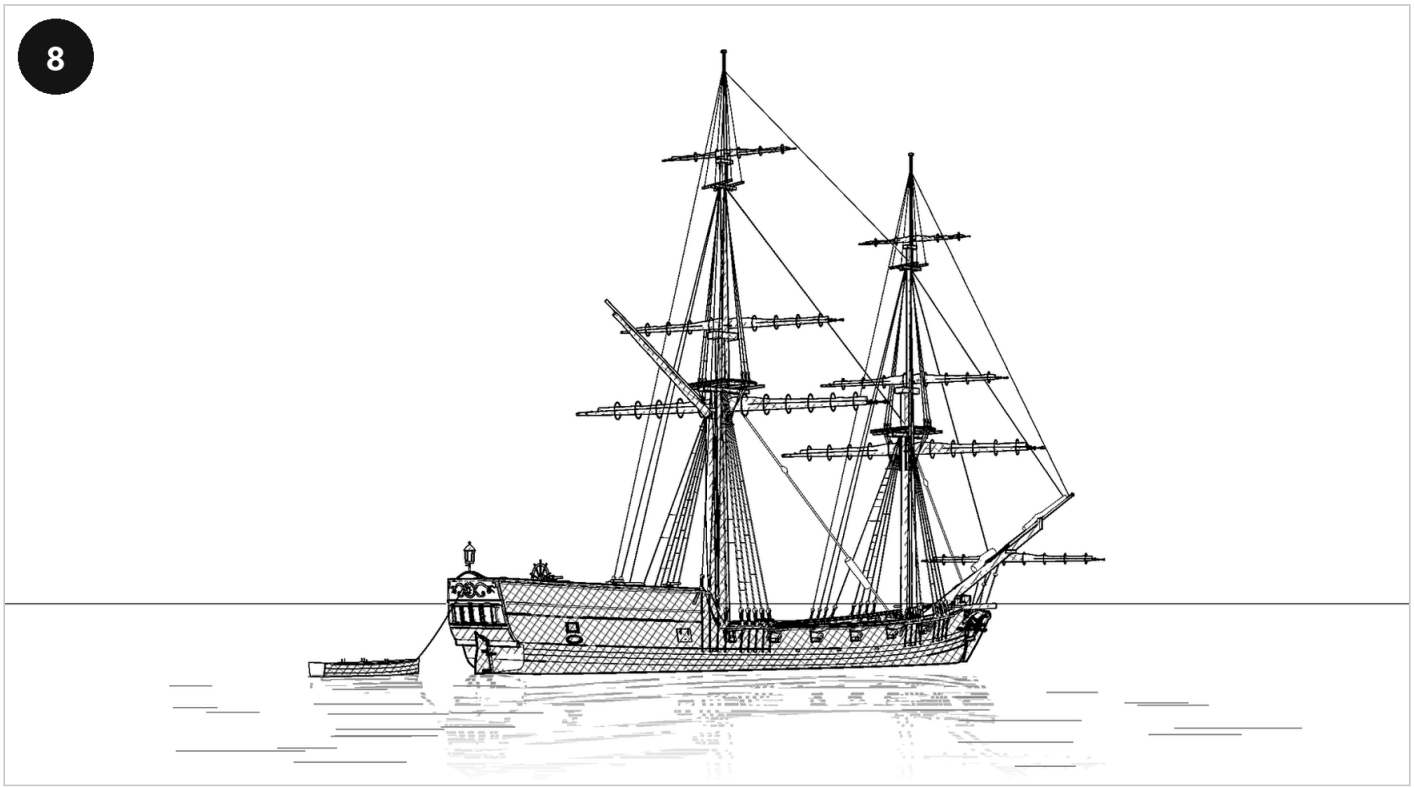


Step 6. Standing rigging. Stays first: long straight lines running forward and down from each mast. Then the shrouds: draw the orange triangles lightly and fill them with lines fanning down to the side of the hull. Ratlines last, and only a few: every third one is plenty.

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Step 7. Hull details. Add the gunport lids along the orange line, with a gun barrel in most of the ports, and the channels: the ledges on the side where the shrouds end. Then the small things: windows and lantern at the stern, anchor at the bow, the steering wheel and the boat on its line behind the ship. The stern has the most detail: take it slowly, window by window.

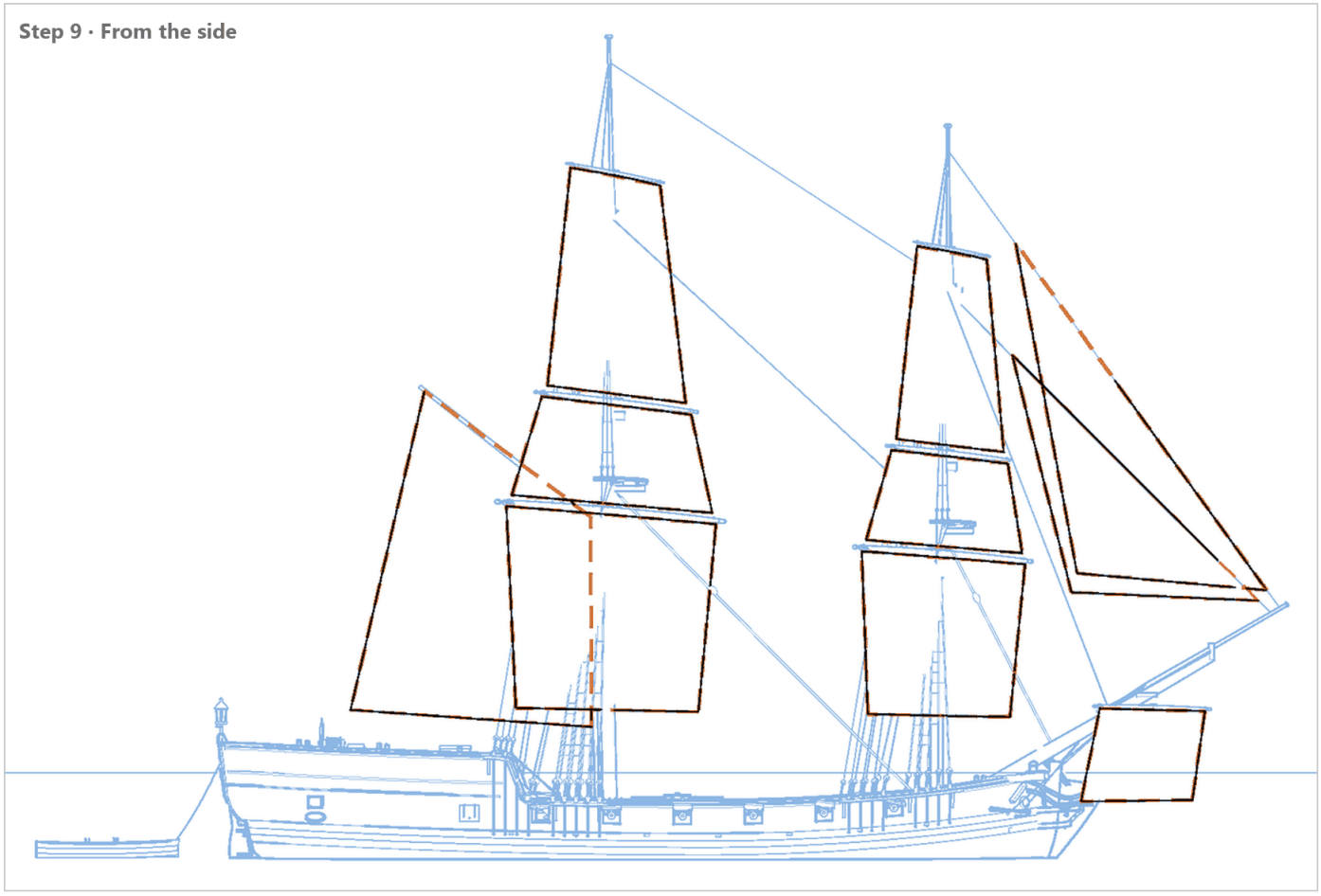


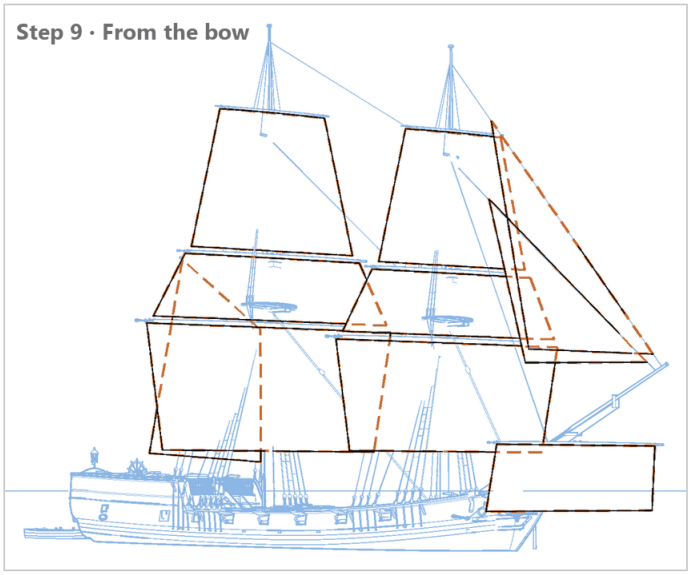
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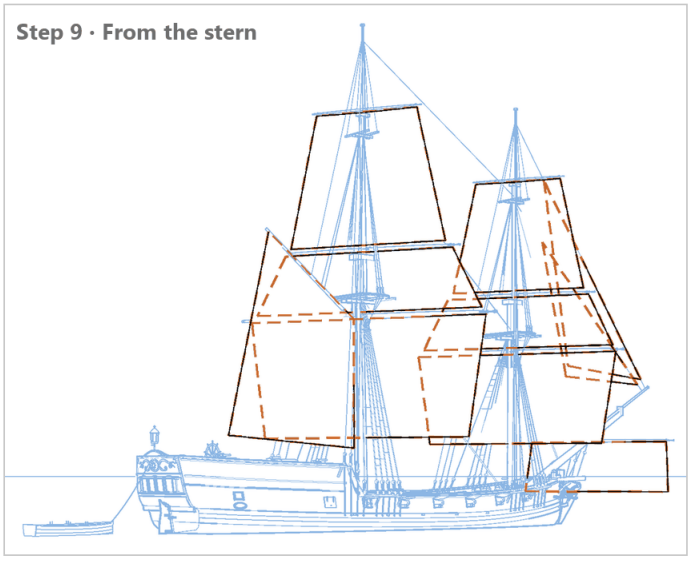
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Bonus challenge: set the sails

In the steps the sails were rolled up on the yards, as on a ship at anchor. Here they are set. Erase the furled sails first, then follow these tips.



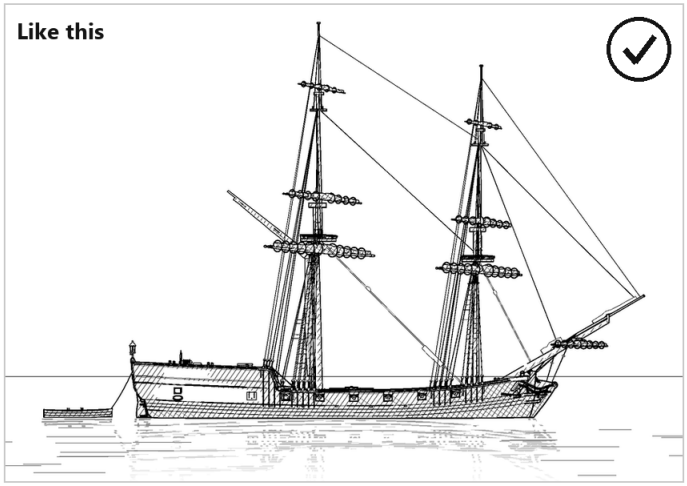


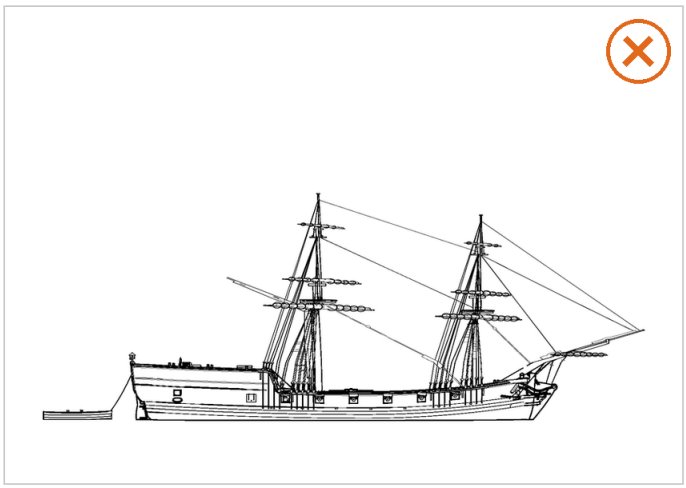


- Draw each sail as a straight four-sided shape first (orange): top corners at the ends of its yard, bottom corners at the ends of the yard below.
- Then curve the edges: the top stays straight along the yard, the sides and the bottom bulge a little, because wind fills the sail.
- Sails hide what is behind them. Erase the masts and ropes behind each sail.
- The gaff sail and the triangular sails at the bow are flat triangles and four-sided shapes along the gaff and the stays.

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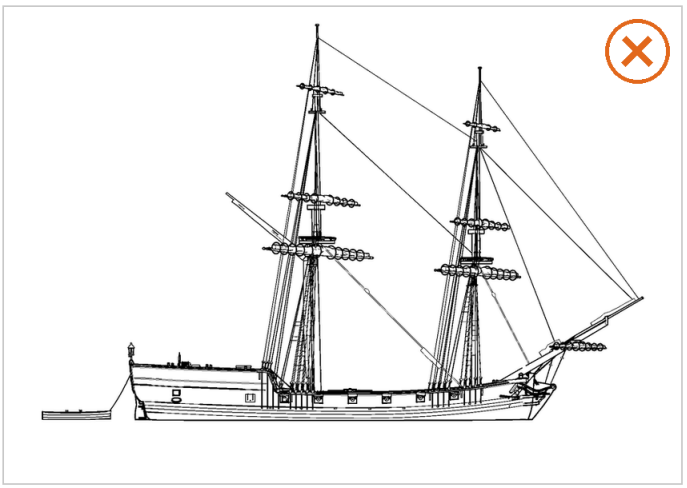
Common mistakes (and how to fix them)





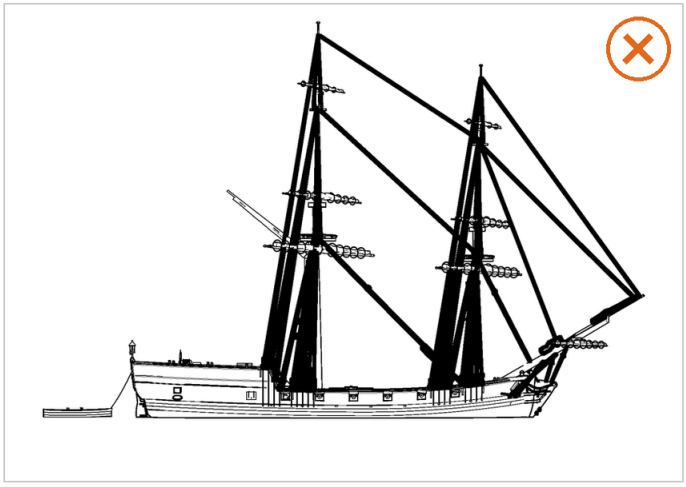
Masts too short

The masts are nearly as tall as the hull is long. Check with your grid before you draw the rigging.



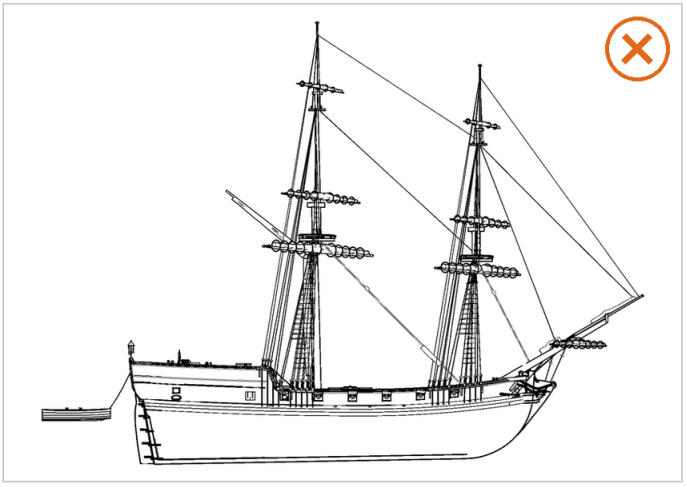
Flat as a board

Only outlines, no light and shadow, no water. Add shading on the side away from the light and a reflection.



Ropes thicker than masts

A rope is about ten times thinner than a mast. Use your sharpest pencil and press lightly.



No waterline

A ship sits in the water: the keel and the bottom are hidden. Start the hull at the waterline.

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Fun facts about our snow

Why the extra thin mast?

Right behind the mainmast stands a thin "snow mast", only about 17 cm thick. The gaff sail runs up and down on it, so the mainmast stays free for its big square sail. This thin mast is what makes the ship a snow.

Ratlines are a ladder

Sailors climbed the shrouds on ratlines: thin ropes tied across them every 33 to 40 cm. Our model has 456 of them.

Ropes are thin

The mainmast is about 49 cm thick. The mainstay, one of the thickest ropes, is only 8 cm, a shroud 5 cm and a ratline 1 cm.

Ten guns

The snow carries 10 cannons, five on each side, and 8 small swivel guns on the rails.

The boat behind

The ship's 5 m boat is towed behind the stern. Before a battle boats were put in the water, so that cannon balls would not smash them into splinters on deck.

How big is it?

The deck is 28 m long and 7.6 m wide, and the ship sits about 3.1 m deep in the water. The top of the mainmast is almost 29 m above the water: nearly as tall as the ship is long.

What do the shrouds do?

Shrouds hold each mast from the sides and a little from behind; stays hold it from the front. Without them the masts would snap or fall over in the first strong wind.

Turning the wheel

The steering wheel is 1.4 m across and has 10 spokes. To swing the rudder from the middle all the way to one side (35 degrees) the helmsman turns the wheel almost twice: 3.65 turns from one side to the other.

Rolled-up sails

In our pictures the sails are furled because the ship is at anchor. Rolling up one big lower sail took 8 to 12 sailors standing on ropes under the yard for 5 to 10 minutes.

All numbers come from the chapters of the construction guide for the reference ship in this repository.

Sources (as cited in the construction guide that goes with this booklet)

D. Steel, The Elements and Practice of Rigging and Seamanship (1794): https://whalesite.org/anthology/1794_Steel_Elements_and_Practice.htm

D. Lever, The Young Sea Officer's Sheet Anchor (1808): https://archive.org/details/youngseaofficers00leve

R. Kenchington, "William Sutherland's Ship, circa 1710", The Northern Mariner (1993): https://cnrs-scrn.org/northern_mariner/vol03/tnm_3_1_1-43.pdf

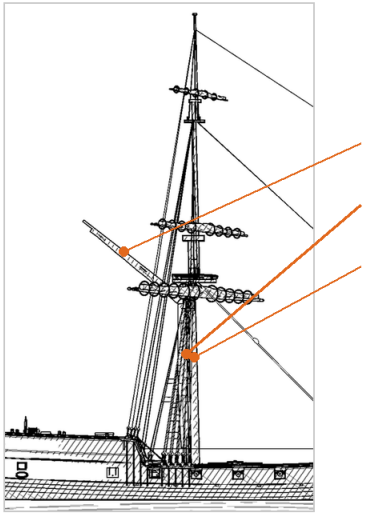
T. Blanckley, A Naval Expositor (1750): https://www.gutenberg.org/ebooks/52902

Royal Museums Greenwich, ship model SLR0416: https://www.rmg.co.uk/collections/objects/rmgc-object-66377

Wikipedia: Snow (ship): https://en.wikipedia.org/wiki/Snow_(ship); Ratlines: https://en.wikipedia.org/wiki/Ratlines; Ship's wheel: https://en.wikipedia.org/wiki/Ship%27s_wheel

Lehman Shipyard, Standing rigging: https://lehmanshipyard.com/2024/04/06/standing-rigging-shrouds-stays/

Polish rigging names: https://potempski.com/rejowe/inne/zagle.htm

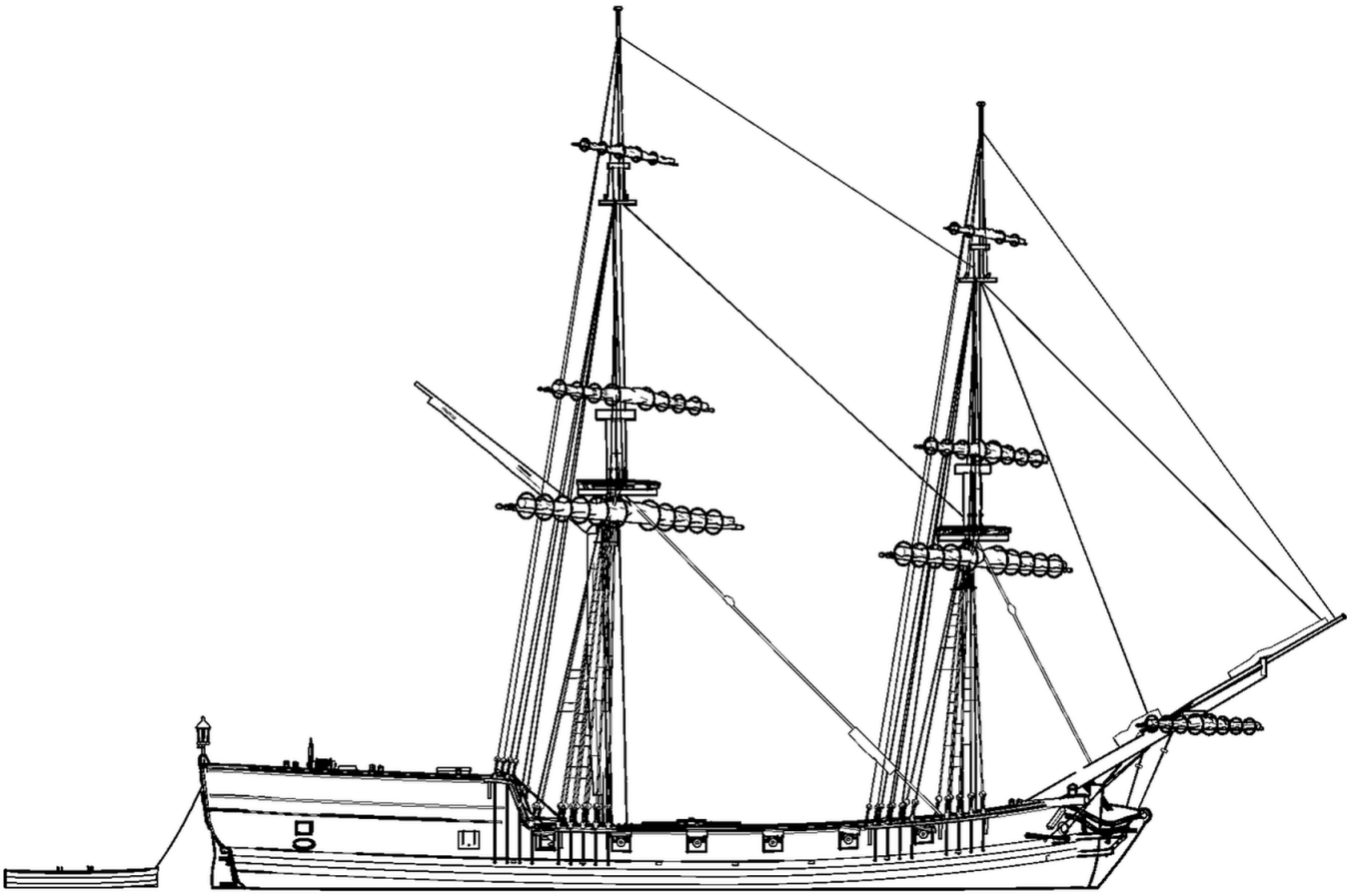


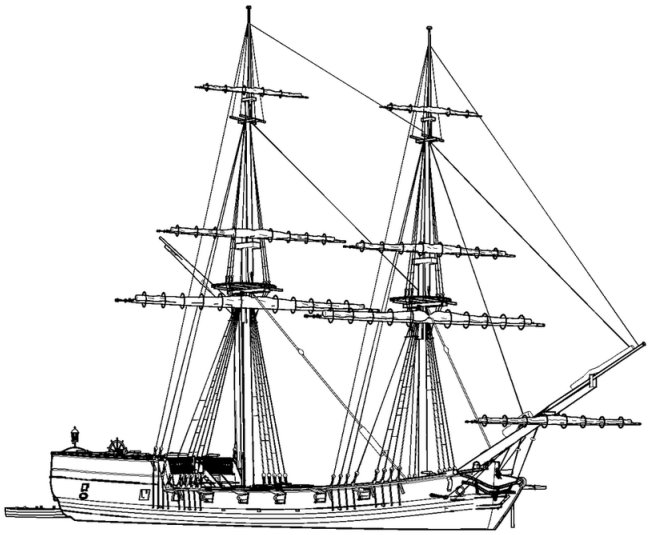
Close-up from the side view: the thin snow mast stands just behind the mainmast. The gaff hangs from it, and the gaff sail is hoisted along this thin mast.

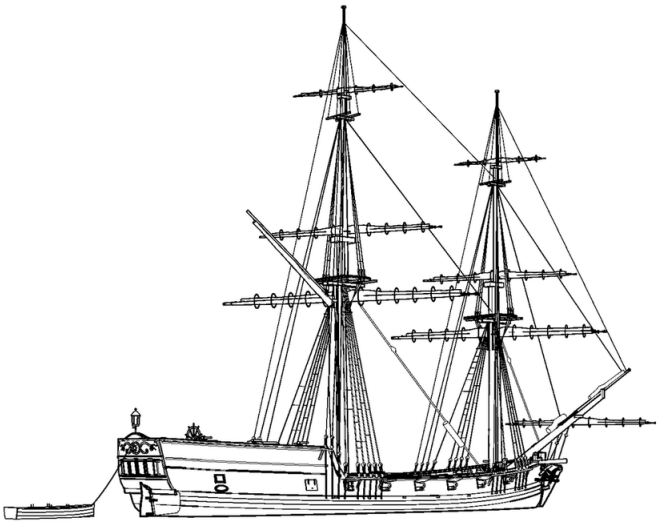
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Tracing and colouring page

Clean outlines of the three views. Put thin paper on top and trace, or print this page and colour it in.







About these pictures: the outlines were rendered from the 3D model with Blender (Freestyle line rendering); construction lines, shading and reflections were computed by Python scripts. For the drawings the square yards are swung slightly (braced) so they show better.

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